Infectious Disease Modelling 10 (2025) 850–874

Contents lists available at ScienceDirect

Infectious Disease Modelling

journal homepage: www.keaipublishing.com/idm

Analysis of the SEIR mean-field model in dynamic networks under intervention

Jiangmin Li [a, b, c], Zhen Jin [a, b, c, *], Ming Tang [d, e]

[a] Complex System Research Center, Shanxi University, Taiyuan, 030006, Shanxi, China
[b] Key Laboratory of Complex Systems and Data Science of Ministry of Education, Shanxi University, Taiyuan, 030006, Shanxi, China
[c] Shanxi Key Laboratory for Mathematical Technology in Complex Systems, Shanxi University, Taiyuan, 030006, Shanxi, China
[d] School of Physics and Electronic Science, East China Normal University, Shanghai, 200241, China
[e] Shanghai Key Laboratory of Multidimensional Information Processing, East China Normal University, Shanghai, 200241, China

ARTICLE INFO

Article history:
Received 17 December 2024
Received in revised form 8 February 2025
Accepted 9 March 2025
Available online 9 April 2025
Handling Editor: Dr. Raluca Eftimie

Keywords:
Tracing isolation
Dynamic network
The final size
Degree distribution

ABSTRACT

For emerging respiratory infectious diseases like COVID-19, non-pharmaceutical interventions such as isolation are crucial for controlling the spread. From the perspective of network transmission, non-pharmaceutical interventions like isolation alter the degree distribution and other topological structures of the network, thereby controlling the spread of the infectious disease. In this paper, we establish a SEIR mean-field propagation dynamics model for the synchronous evolution of dynamic networks caused by propagation and tracing isolation. We employ the reducing-dimension method to convert the mean-field model in networks into an equivalent and simpler low-dimension model, and then calculate the exact expression of the final size. In addition, we get the differential equations of the degree distribution over time in dynamic networks under tracing isolation and the relationships between the first and second moment of the dynamic network. While the degree of a node remains constant regardless of its state in many previous studies, this paper takes into account that the degree of each node changes over time whatever its state under the disease spread and intervention measures.

© 2025 The Authors. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

1. Introduction

In recent years, the study of complex network disease propagation dynamics model is a hot topic and there have been a large number of research results. Among the representative works, Newman et al. proposed some methods about probability generating function of distribution for network structure in literature (Strogatz, 2001; Newman et al., 2001; Newman, 2009; Newman, 2002), Pastor et al. applied network theory to the modeling of disease propagation (Boguná & Pastor-Satorras, 2002; Pastor-Satorras and Vespignani, 2001a; Pastor-Satorras and Vespignani, 2001b; Pastor-Satorras & Vespignani, 2002; Moreno et al., 2002), and Volz et al. established a network SIR disease propagation model based on edges (Miller, 2011; Volz, 2008).

* Corresponding author. Complex System Research Center, Shanxi University, Taiyuan, 030006, Shanxi, China.
E-mail address: jinzhn@263.net (Z. Jin).
Peer review under the responsibility of KeAi Communications Co., Ltd.

https://doi.org/10.1016/j.idm.2025.03.002
2468-0427/© 2025 The Authors. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

The Pastor model only includes the changes of nodes with different degrees, but the spread of disease in the network is completely through the edge of nodes. Therefore, learning from the idea of the moment-closed model, Volz proposed a network disease propagation model in 2008 that considered network edges (Volz, 2008). And it solves the problem that the Pastor model overestimates the disease transmission threshold. In addition, Miller simplified the Volz model mathematically in 2011 and gave a more concise and clearer equivalent model (Miller, 2011). Based on the Miller and Pastor models, it is obviously more meaningful to study models with tracing isolation measures. Zhang et al. introduces a SEIS model on a multi-layer network, deriving its steady-state equations and epidemic threshold, and reveals the cumulative effect on the infected as well as the influence of minimal mobile crowds on epidemic prevalence at the steady state (Zhang et al., 2023). Besides, Chen et al. investigates a coupled model for resource allocation and epidemic spreading, which considers both the allocation decisions and delay behavior of individuals with limited resources. The study analyzes how resource allocation delays impact epidemic spreading and identifies the optimal delay for maximizing the epidemic threshold (Chen et al., 2025). In a word, these works offer valuable insights into epidemic dynamics and control strategies. In 2020, Luo et al. reduced the Pastor model to the more efficient lower-dimensional Volz model and also explored disease transmission models with interventions (Luo & Jin, 2020). This work focused more on the impact of isolating high-degree nodes on the final size and duration of the disease in network transmission.

Since the outbreak of COVID-19, tracing isolation on the network has played a pivotal role. Okolie et al. considered the contact tracing of the random SIR model on the random tree and the configuration model, extended the existing uniform mixed population contact tracing theory to the tree. Besides, they established a tree dynamic model and an approximate mean-field equation (Okolie & Müller, 2020). Kojaku et al. studied the effectiveness of backward contact tracing in the network (Kojaku et al., 2021). Also studied forward and backward contact tracing are Endo et al. Using a simple branching process model on the network, they combined backward contact tracing with traditional forward contact tracing to control this disease, estimated the size of a typical cluster that can be achieved through backward contact tracing, and simulated the effectiveness of combining backward tracing with traditional forward tracing (Endo et al., 2020). Ball et al. studied contact tracing using branching process (Ball et al., 2007; Ball et al., 2008; Ball et al., 2011). In fact, the branching process method can only be used to study early disease transmission, whereas the mean-field dynamics model is more suitable for studying large-scale and longtime disease transmission. Therefore, it is very meaningful to use the network mean-field dynamics model to study the transmission dynamics of diseases under tracing isolation measures.

There are also numerous studies on dynamic networks that explore the interplay between network growth and contraction, as well as the process of network contraction. In the landmark paper by Barabási and Albert (Barabási, 1999), it was shown that networks evolving through growth processes with preferential attachment naturally tend towards scale-free structures, which provided a framework to understand the emergence of highly heterogeneous networks in many real-world systems. Tishby I et al. (Tishby et al., 2019; Tishby et al., 2020) explored network contraction processes, where networks subjected to random, preferential, or propagating node deletions evolve toward an ER structure with a Poisson degree distribution. Budnick B et al. (Budnick et al., 2022; Budnick et al., 2025) examined the interplay between network growth and contraction, identifying phase transitions and steady states.

The rest of this paper is organized as follows. In section 2, we give the equivalent low-dimensional model of the SEIR mean-field model and analyze its corresponding dynamical behavior. And based on the tracing isolation policy, the differential equation of the degree distribution related to time t is deduced and the SEIR mean-field model of dynamic network under the tracing isolation is given. The same dimensionality reduction and dynamics analysis process is carried out. In section 3, we estimate the relevant parameters in the propagation process and gain the corresponding fitting result with the actual data. Finally, we make a brief discussion and conclusion in the last section.

2. Methods

2.1. SEIR mean-field model

When constructing infectious disease compartment models, people are usually divided into susceptible (S), exposed (E), infected (I) and recovered (R). In classical infectious disease models, there are several structures to consider, such as SIR, SIS, SEIR models and so on. The choice of model really depends on the transmission mechanism of the infectious disease. Following infection with COVID-19, there exists a significant exposed period before infecting others. And the SEIR model has recently been widely employed to describe the COVID-19 infection (Mielke & Christiansen, 2025; Nava et al., 2020; Sabbatino et al., 2025). Therefore, we employ the SEIR mean-field model to accurately capture this dynamic.

2.1.1. The introduction of SEIR mean-field model

Considering the degree-uncorrelated and state-uncorrelated network with degree distribution $p(k)$, without considering clustering, birth and death, the total population is constant.

The probability generating function of distribution $p(k)$ is defined as $G_0(x) = \sum_{k=0}^{\infty} p(k)x^k$ with $x \in [0, 1]$. The average degree of this network (the first moment of the distribution) is $\langle k \rangle = \sum_{k=0}^{\infty} kp(k) = G_0'(1)$ and the second moment is $\langle k^2 \rangle = \sum_{k=0}^{\infty} k^2 p(k) = G_0'(1) + G_0''(1)$. The relevant definitions of $G_1(x)$ are in Appendix A.

Firstly, let us recall the classical SEIR mean-field model. Given the degree distribution $p(k)$, the classical SEIR mean-field model (Jin et al., 2014) is

$$\frac{dS_k(t)}{dt} = -\beta k\phi_I(t)S_k(t), \tag{1a}$$

$$\frac{dE_k(t)}{dt} = \beta k\phi_I(t)S_k(t) - \epsilon E_k(t), \tag{1b}$$

$$\frac{dI_k(t)}{dt} = \epsilon E_k(t) - \gamma I_k(t), \tag{1c}$$

$$\frac{dR_k(t)}{dt} = \gamma I_k(t), \tag{1d}$$

where $S_k(t)$, $E_k(t)$, $I_k(t)$ and $R_k(t)$ represent the relative densities of susceptible, exposed, infected and recovered nodes with degree k at time t, respectively, satisfying the normalization condition $S_k(t) + E_k(t) + I_k(t) + R_k(t) = 1$.

One susceptible individual is infected by his/her infected neighbors at the transmission rate β, an exposed individual becomes infectious at rate ϵ and infected individuals recover at rate γ. $\phi_I(t)$ represents the probability that any chosen edge is connected to an infected node at time t,

$$\phi_I(t) = \frac{\sum_{k=0}^{\infty} kp(k)I_k(t)}{\langle k\rangle}. \tag{2}$$

Similarly, $\phi_S(t)$, $\phi_E(t)$ and $\phi_R(t)$ represent the probabilities that any chosen edge is connected to a susceptible, exposed and recovered node at time t, respectively,

$$\phi_S(t) = \frac{\sum_{k=0}^{\infty} kp(k)S_k(t)}{\langle k\rangle}, \phi_E(t) = \frac{\sum_{k=0}^{\infty} kp(k)E_k(t)}{\langle k\rangle}, \phi_R(t) = \frac{\sum_{k=0}^{\infty} kp(k)R_k(t)}{\langle k\rangle}. \tag{3}$$

Then, the densities of susceptible, exposed, infected and recovered nodes in network, are respectively expressed as

$$\begin{aligned} S(t) &= \sum_{k=0}^{\infty} p(k)S_k(t), E(t) = \sum_{k=0}^{\infty} p(k)E_k(t), \\ I(t) &= \sum_{k=0}^{\infty} p(k)I_k(t), R(t) = \sum_{k=0}^{\infty} p(k)R_k(t). \end{aligned} \tag{4}$$

2.1.2. The equivalent low-dimensional model

In this section, the derivation method similar to the Volz model (Volz, 2008) to the Miller model (Miller, 2011) is used to transform the SEIR mean-field model (1) into an equivalent low-dimensional model. Firstly, solving Eq. (1a) yields

$$S_k(t) = S_k(0)\exp\left[-\beta k\int_0^t \phi_I(\tau)d\tau\right] = S_k(0)\theta^k(t), \tag{5}$$

where

$$\theta(t) = \exp\left[-\beta\int_0^t \phi_I(\tau)d\tau\right] = \exp\left[-\frac{\beta}{\gamma}\phi_R(t)\right] \tag{6}$$

represents the probability that any given edge emerging from an initial susceptible node remains uninfected until time t. The initial conditions are denoted by $S_k(0)$, $E_k(0)$, $I_k(0)$ and $R_k(0)$, respectively. At the onset of epidemic, almost all the individuals are susceptible in the population with few exposed, infected individuals and no recovery individuals, so we assume $S_k(0) \approx 1$, $E_k(0) \approx 0$, $I_k(0) \approx 0$ and $R_k(0) = 0$. Furthermore, $S_k(t) = \theta^k(t)$ and

$$S(t) = \sum_{k=0}^{\infty} p(k)\theta^k(t) = G_0(\theta(t)), \phi_S(t) = \frac{\theta(t)G_0'(\theta(t))}{\langle k\rangle}. \tag{7}$$

Then, differentiating $\theta(t)$ in Eq. (6) with respect to time t leads to

$$\frac{d\theta(t)}{dt} = -\beta\theta(t)\phi_I(t). \tag{8}$$

In addition, according to Eqs. (1c), (2) and (3), we get the evolution equation of $\phi_I(t)$ with time t,

$$\frac{d\phi_I(t)}{dt} = \epsilon\phi_E(t) - \gamma\phi_I(t). \tag{9}$$

Similarly, according to Eqs. (1b), (2) and (3), we get the evolution equation of $\phi_E(t)$ with time t,

$$\frac{d\phi_E(t)}{dt} = \beta\phi_I(t)\frac{\theta^2(t)G_0''(\theta(t)) + \theta(t)G_0'(\theta(t))}{\langle k\rangle} - \epsilon\phi_E(t). \tag{10}$$

With the above derivation, the equivalent reduced-dimension model is

$$\begin{aligned}
&\frac{d\theta(t)}{dt} = -\beta\theta(t)\phi_I(t),\\
&\frac{d\phi_E(t)}{dt} = \beta\phi_I(t)\frac{\theta^2(t)G_0''(\theta(t)) + \theta(t)G_0'(\theta(t))}{\langle k\rangle} - \epsilon\phi_E(t),\\
&\frac{d\phi_I(t)}{dt} = \epsilon\phi_E(t) - \gamma\phi_I(t),\\
&\frac{dI(t)}{dt} = \epsilon E(t) - \gamma I(t),\\
&\frac{dR(t)}{dt} = \gamma I(t),\\
&S(t) = G_0(\theta(t)),\\
&E(t) = 1 - S(t) - I(t) - R(t).
\end{aligned} \tag{11}$$

According to the third formula in Eq. (3), we have

$$\frac{d\phi_R(t)}{dt} = \sum_{k=0}^{\infty}\frac{kp(k)}{\langle k\rangle}\frac{dR_k(t)}{dt} = \gamma\phi_I(t). \tag{12}$$

Based on Eq. (6), $\int_0^t \phi_I(\tau)d\tau = -\frac{1}{\beta}\ln\theta(t)$. So

$$\phi_R(t) = -\frac{\gamma}{\beta}\ln\theta(t). \tag{13}$$

Since $\phi_S(t) + \phi_E(t) + \phi_I(t) + \phi_R(t) = 1$, then

$$\frac{d\phi_I(t)}{dt} + (\epsilon+\gamma)\phi_I(t) = \epsilon\left[1 - \frac{\theta(t)G_0'(\theta(t))}{\langle k\rangle} + \frac{\gamma}{\beta}\ln\theta(t)\right]. \tag{14}$$

According to the first formula in Eq. (11), we have

$$\begin{aligned}
&\phi_I(t) = \frac{\theta'(t)}{-\beta\theta(t)},\\
&\phi_I'(t) = \frac{\theta''(t)\theta(t) - [\theta'(t)]^2}{-\beta\theta^2(t)}.
\end{aligned} \tag{15}$$

Then Eq. (14) can also be written as follows

$$\theta''(t)\theta(t) - [\theta'(t)]^2 + (\epsilon+\gamma)\theta(t)\theta'(t) = -\beta\epsilon\theta^2(t)\left[1 - \frac{\theta(t)G_0'(\theta(t))}{\langle k\rangle} + \frac{\gamma}{\beta}\ln\theta(t)\right]. \tag{16}$$

Integrating both sides of the equation with respect to time t yields

$$\phi_I(t) + \frac{(\epsilon+\gamma)}{\gamma}\phi_R(t) = \int_0^t \epsilon\left[1 - \frac{\theta(\tau)G_0'(\theta(\tau))}{\langle k\rangle} + \frac{\gamma}{\beta}\ln\theta(\tau)\right]d\tau. \tag{17}$$

Arrange the above formula to get

$$\phi_I(t) = \frac{(\epsilon+\gamma)}{\beta}\ln\theta(t) + \epsilon t - \frac{\epsilon}{\langle k\rangle}\int_0^t \theta(\tau)G_0'(\theta(\tau))d\tau + \frac{\epsilon\gamma}{\beta}\int_0^t \ln\theta(\tau)d\tau. \tag{18}$$

Now ϕ_I can be expressed in terms of θ, the solution for Eq. (1b) can be obtained based on it.

$$E_k(t) = \beta k\exp(-\epsilon t)\int_0^t \phi_I(\tau)\theta^k(\tau)\exp(\epsilon\tau)d\tau. \tag{19}$$

Then

$$E(t) = \beta\exp(-\epsilon t)\int_0^t \phi_I(\tau)\theta(\tau)G_0'(\theta(\tau))\exp(\epsilon\tau)d\tau. \tag{20}$$

In terms of the fourth formula in Eq. (11), we can also solve for

$$I(t) = \epsilon\exp(-\gamma t)\int_0^t \exp(\gamma\tau)E(\tau)d\tau. \tag{21}$$

And $R(t) = 1 - S(t) - E(t) - I(t)$, this means that $S(t)$, $E(t)$, $I(t)$, $R(t)$ can always be represented by $\theta(t)$.

2.1.3. Basic reproduction number and final size

In this section, we start to study the basic reproduction number and final size of model (1). The basic reproduction number R_0, also called the basic reproductive number or the basic reproductive ratio, is the single most important parameter in epidemic modeling. It measures the average number of secondary infections caused by a single infectious individual in an entirely susceptible population during the mean infectious period (Ma et al., 2004). If $R_0 < 1$, then an epidemic will not occur; if $R_0 > 1$, an epidemic will occur.

Define the auxiliary function as follows

$$\psi_1(t) = \int_0^t \phi_I(\tau)d\tau = \frac{\sum_{k=0}^{\infty} kp(k)R_k(t)}{\gamma\langle k\rangle}. \tag{22}$$

Easy to get $0 < \psi_1(t) < 1/\gamma$ and $\frac{d\psi_1(t)}{dt} = \frac{\sum_{k=0}^{\infty} kp(k)I_k(t)}{\langle k\rangle} = \phi_I(t)$. Multiplying both sides of Eq. (1c) by $\frac{kp(k)}{\langle k\rangle}$, summing up k and and calculating the integral yield

$$\frac{d\psi_1(t)}{dt} = \epsilon\psi_2(t) - \gamma\psi_1(t) \tag{23}$$

where

$$\psi_2(t) = \int_0^t \phi_E(\tau)d\tau. \tag{24}$$

Similarly, $\frac{d\psi_2(t)}{dt} = \phi_E(t) = \frac{\sum_{k=0}^{\infty} kp(k)E_k(t)}{\langle k\rangle}$. Multiplying both sides of Eq. (1b) by $\frac{kp(k)}{\langle k\rangle}$, summing up k and and calculating the integral yield

$$\frac{d\psi_2(t)}{dt} = 1 - G_1(-\beta\psi_1(t)) - \epsilon\psi_2(t). \tag{25}$$

Next, solve the equilibrium of the following system.

$$\begin{cases} \dfrac{d\psi_1(t)}{dt} = \epsilon\psi_2(t) - \gamma\psi_1(t) \triangleq P(\psi_1(t), \psi_2(t)), \\ \dfrac{d\psi_2(t)}{dt} = 1 - G_1(-\beta\psi_1(t)) - \epsilon\psi_2(t) \triangleq Q(\psi_1(t), \psi_2(t)), \end{cases} \tag{26}$$

Let $\frac{d\psi_1(t)}{dt} = 0, \frac{d\psi_2(t)}{dt} = 0$, we can acquire the following equation about $\psi_1(t)$

$$\frac{1}{\gamma}[1 - G_1(-\beta\psi_1(t))] = \psi_1(t). \tag{27}$$

Obviously, $\psi_1(t) = 0$ is a solution of Eq. (27), which corresponds to no outbreak of the infectious diseases. Let $f(\psi_1(t)) = \frac{1}{\gamma}[1 - G_1(-\beta\psi_1(t))]$, based on Eq. (22), $0 < f(1/\gamma) < 1/\gamma$ is obviously established. According to the existence of solution, the following equation must be satisfied

$$\frac{d}{d\psi_1(t)} f(\psi_1(t))\Big|_{\psi_1(t)=0} \geq 1, \tag{28}$$

i.e.,

$$R_0 = \frac{\beta}{\gamma}\frac{\langle k^2\rangle}{\langle k\rangle}. \tag{29}$$

Therefore Eq. (27) has an unique solution $\psi_1(t) = \psi_1^*$ in the interval (0,1), i.e.,

$$\frac{1}{\gamma}\left[1 - G_1(-\beta\psi_1^*)\right] = \psi_1^*, \tag{30}$$

which means the disease breaks out.

As a consequence, for system (26), there are two solutions

$$\begin{cases} \psi_1(t) = 0 \\ \psi_2(t) = 0 \end{cases} \quad \begin{cases} \psi_1(t) = \psi_1^* \\ \psi_2(t) = \psi_2^* \end{cases} \tag{31}$$

where ψ_1^* satisfies $\frac{1}{\gamma}\left[1 - G_1(-\beta\psi_1^*)\right] = \psi_1^*$, and ψ_2^* can be solved by this formula and system (26).

Since $\frac{d}{dt}[S_k(t) + E_k(t)] = -\epsilon E_k(t)$, then $S_k(t) + E_k(t)$ is monotonically decreasing and there exists a lower bound 0. So $\lim_{t\to\infty} E_k(t) = 0$ is established. Similarly, $\lim_{t\to\infty} I_k(t) = 0$. Therefore, $\lim_{t\to\infty} S_k(t) + R_k(t) = 1$, $\lim_{t\to\infty} E_k(t) = 0$, $\lim_{t\to\infty} I_k(t) = 0$ are finally satisfied.

For $\phi_E(t)$, we have

$$\lim_{t\to\infty}\phi_E(t) = \lim_{t\to\infty}\frac{\sum_{k=0}^{\infty} kp(k)E_k(t)}{\langle k\rangle} = 0. \tag{32}$$

In other words, there exists $M_E > 0$ such that $\phi_E(t) \leq M_E$, then we can take a sufficiently large $T_E > 0$ such that $\psi_2(t) = \int_0^t \phi_E(\tau)d\tau \leq \int_0^{T_E} M_E d\tau = M_E T_E$. As a result, we get a positive invariant set $\Omega = \{(\psi_1, \psi_2) | 0 \leq \psi_1 \leq 1/\gamma, 0 \leq \psi_2 \leq M_E T_E\}$.

Considering that the Jacobian matrix corresponding to system (26) is

$$J = \begin{pmatrix} -\gamma & \epsilon \\ \beta\displaystyle\sum_{k=0}^{\infty}\frac{k^2 p(k)}{\langle k\rangle}\exp[-\beta k\psi_1(t)] & -\epsilon \end{pmatrix}. \tag{33}$$

The first set of solutions mentioned above is the disease-free equilibrium and the corresponding Jacobian matrix is

$$J = \begin{pmatrix} -\gamma & \epsilon \\ \beta\dfrac{\langle k^2\rangle}{\langle k\rangle} & -\epsilon \end{pmatrix}, \tag{34}$$

the related characteristic equation is $\lambda^2 + (\gamma + \epsilon)\lambda + \gamma\epsilon - \frac{\beta\epsilon\langle k^2\rangle}{\langle k\rangle} = 0$. Then $\lambda_1 + \lambda_2 = -(\gamma + \epsilon) < 0, \lambda_1\lambda_2 = \gamma\epsilon - \frac{\beta\epsilon\langle k^2\rangle}{\langle k\rangle}$. So when $R_0 < 1$, the disease-free equilibrium is locally asymptotically stable; when $R_0 > 1$, the disease-free equilibrium is unstable. In positive invariant set Ω there is only one equilibrium when $R_0 < 1$, so there is no closed orbit. Then the disease-free equilibrium is globally asymptotically stable in positive invariant set Ω.

The following proves that the positive equilibrium corresponding to the second set of solutions mentioned above is globally stable when $R_0 > 1$. Considering the Jacobian matrix

$$J = \begin{pmatrix} -\gamma & \epsilon \\ \beta\sum_{k=0}^{\infty}\frac{k^2p(k)}{\langle k\rangle}\exp(-\beta k\psi_1^*) & -\epsilon \end{pmatrix} \tag{35}$$

corresponding to the solution, the related characteristic equation is

$$\lambda^2 + (\gamma+\epsilon)\lambda + \gamma\epsilon - \epsilon\beta\sum_{k=0}^{\infty}\frac{k^2p(k)}{\langle k\rangle}\exp(-\beta k\psi_1^*) = 0. \tag{36}$$

Then $\lambda_1 + \lambda_2 = -(\gamma + \epsilon) < 0, \lambda_1\lambda_2 = \gamma\epsilon - \epsilon\beta\sum_{k=0}^{\infty}\frac{k^2p(k)}{\langle k\rangle}\exp(-\beta k\psi_1^*)$. Next, judge the symbol of $\lambda_1\lambda_2$. Derivation of ψ_1^* on both sides of Eq. (27), we have $\sum_{k=0}^{\infty}\frac{k^2p(k)}{\langle k\rangle}\exp(-\beta k\psi_1^*) = \gamma/\beta$, i.e., $\lambda_1\lambda_2 = 0$. Consequently, when $R_0 > 1$ the positive equilibrium is locally stable. The Bendixson-Dulac Theorem (Jordan & Smith, 2007) is used to discuss the acyclicity of system (26). Let $B(\psi_1, \psi_2) = \psi_1^a\psi_2^b$, where a and b are undetermined constants, then

$$\begin{aligned} D &= \frac{\partial(BP)}{\partial\psi_1} + \frac{\partial(BQ)}{\partial\psi_2} \\ &= \psi_1^{a-1}\psi_2^{b-1}\left\{b\psi_1[1 - G_1(-\beta\psi_1) - \epsilon\psi_2] + a\epsilon\psi_2^2 - (a\gamma + \gamma + \epsilon)\psi_1\psi_2\right\}. \end{aligned} \tag{37}$$

Let $a = -1$, $b = 0$, then

$$D = \psi_1^{-2}\psi_2^{-1}(-\epsilon\psi_2^2 - \epsilon\psi_1\psi_2) = -\epsilon\psi_1^{-2}\psi_2 - \epsilon\psi_1^{-1} = -\epsilon\psi_1^{-1}(\psi_1^{-1}\psi_2 + 1) \le 0. \tag{38}$$

By selecting Dulac function $B(\psi_1, \psi_2) = \psi_1^{-1}$, $D \le 0$ can be established in the whole plane. According to Bendixson-Dulac theorem, there is no closed orbit in positive invariant set Ω for system (26), so the positive equilibrium is globally stable. Therefore we have the final conclusion.

Theorem 1. *For system* (26)*, when $R_0 < 1$ the disease-free equilibrium is globally asymptotically stable in Ω; when $R_0 > 1$, the disease-free equilibrium becomes unstable and a unique positive equilibrium is globally stable in Ω.*

If $R_0 > 1$, epidemic ends with $\psi_1(t) \to \psi_1^*$ and $\theta(t) = \exp[-\beta\psi_1(t)] \to \exp(-\beta\psi_1^*) \triangleq \theta^*$ for $t \to \infty$. In fact, $\theta*$ is the constant probability of one randomly chosen edge remains uninfected by the end of epidemic. By Eq. (4), the final density of susceptible nodes in network is

$$S^\infty \triangleq S(\infty) = \sum_{k=0}^{\infty}p(k)\exp(-\beta k\psi_1^*) = G_0(-\beta\psi_1^*). \tag{39}$$

Furthermore, the final size of model (1) is

$$R^\infty \triangleq 1 - S(\infty) = 1 - G_0(-\beta\psi_1^*). \tag{40}$$

2.2. The dynamic network epidemic model under intervention

This section will discuss the change of the degree distribution of the dynamic network over time and establish a mean-field dynamics model under this dynamic network. In addition, based on the real data of COVID-19, the MCMC (Markov Chain Monte Carlo) algorithm will be used to perform some parameter estimation and data fitting under intervention.

2.2.1. SEIR mean-field model on dynamic network

For the tracing isolation, we consider the impact of isolating infected nodes, close contact nodes, and indirect contact nodes after the infected is found in the nucleic acid detection on the network degree distribution COVID-19 prevention and control program, (2020). The network before isolation is exactly the same as above.

We suppose that the infected nodes are found at time T and then the infected nodes, close contact nodes and the indirect contact nodes are isolated. At the same time we assume that for the infected node, the probability of its first-order neighbors (close contact nodes) being found is a_1, the probability of its second-order neighbors (indirect contact nodes) being found is a_2 and the probability of its third-order neighbors being found is a_3. Where the first-order neighbors of a node are nodes connected to the node, the second-order neighbors of a node are nodes connected to the first-order neighbors of the node, similarly, the n-order neighbors of a node are nodes connected to the $(n-1)$-order neighbors of the node.

The degree distribution after time T is always changing over time. We can first analyze the change of degree distribution after the isolation of time T (in Appendix B).

Similar to the discussion of the network degree distribution after time T in Appendix B, we can discuss the network degree distribution $p_k(t)$ at time t. The probability generating function is denoted by

$$G(t,x)=\sum_{k=0}^{\infty}p_k(t)x^k \tag{41}$$

with $x \in [0,1]$. The average degree of this network and the second moment of the distribution are respectively

$$\langle k\rangle_t=\sum_{k=0}^{\infty}kp_k(t)=G'_x(t,1),\langle k^2\rangle_t=\sum_{k=0}^{\infty}k^2p_k(t)=G_x\prime\prime(t,1)+G'_x(t,1). \tag{42}$$

Let the average number of the first-order, second-order and d-order neighbors (Newman et al., 2001) of a node at time t be

$$C_1(t)=\langle k\rangle_t, C_2(t)=\langle k^2\rangle_t-\langle k\rangle_t, C_d(t)=\left[\frac{C_2(t)}{C_1(t)}\right]^{d-1}C_1(t). \tag{43}$$

The density of the indirect contact nodes and the third-order neighbor nodes found in the whole network at time t are respectively

$$\begin{aligned} &a_I a_2 C_2(t)I(t)\triangleq M(t),\\ &a_I a_3 C_3(t)I(t)\triangleq L(t). \end{aligned} \tag{44}$$

In a word, the density of isolated nodes in the whole network at time t is

$$a_I I(t)+a_I a_1 C_1(t)I(t)+a_I a_2 C_2(t)I(t)=a_I I(t)[1+a_1C_1(t)+a_2C_2(t)]\triangleq Q(t). \tag{45}$$

Similarly, we can define the probability of an indirect contact node connecting to a third-order neighbor with degree k at time t as $q_k(t)\triangleq M(t)\frac{kp_k(t)}{\langle k\rangle_t}$.

After some similar derivations in B, the degree distribution can be estimated as follows

$$p_k(t+\Delta t)\approx\frac{p_k(t)-p_k(t)Q(t)\Delta t-q_k(t)L(t)\Delta t+q_{k+1}(t)L(t)\Delta t}{1-Q(t)\Delta t}. \tag{46}$$

Subtracting the degree distribution at time $t+\Delta t$ and time t

$$\begin{aligned} p_k(t+t)-p_k(t)&\approx\frac{p_k(t)-p_k(t)Q(t)\Delta t-q_k(t)L(t)\Delta t+q_{k+1}(t)L(t)\Delta t}{1-Q(t)\Delta t}-p_k(t)\\ &=\frac{q_{k+1}(t)L(t)\Delta t-q_k(t)L(t)\Delta t}{1-Q(t)\Delta t}. \end{aligned} \tag{47}$$

Then the equation can be written as

$$\begin{aligned}\frac{dp_k(t)}{dt} &= \lim_{\Delta t\to 0}\frac{p_k(t+\Delta t)-p_k(t)}{\Delta t}\\ &= \lim_{\Delta t\to 0}\frac{q_{k+1}(t)L(t)-q_k(t)L(t)}{1-Q(t)\Delta t}\\ &= [(k+1)p_{k+1}(t)-kp_k(t)]\frac{a_2a_3\left(\left\langle k^2\right\rangle_t-\langle k\rangle_t\right)^3}{\left(\langle k\rangle_t\right)^2}[a_II(t)]^2.\end{aligned} \tag{48}$$

Multiplying both sides of this equation by x^k and summing k from 0 to ∞, then

$$\begin{aligned}G_t^{'}(t,x) &= \sum_{k=0}^{\infty}\frac{dp_k(t)}{dt}x^k\\ &= \frac{a_2a_3\left(\left\langle k^2\right\rangle_t-\langle k\rangle_t\right)^3}{\left(\langle k\rangle_t\right)^2}[a_II(t)]^2\sum_{k=0}^{\infty}[(k+1)p_{k+1}(t)-kp_k(t)]x^k\\ &= \frac{a_2a_3\left(\left\langle k^2\right\rangle_t-\langle k\rangle_t\right)^3[a_II(t)]^2}{\left(\langle k\rangle_t\right)^2}G_x^{'}(t,x)(1-x).\end{aligned} \tag{49}$$

By multiplying k on both sides of Eq. (49) and summing k from 0 to ∞, we have

$$\begin{aligned}\frac{d\langle k\rangle_t}{dt} &= \frac{d\sum\limits_{k=0}^{\infty}kp_k(t)}{dt}\\ &= \sum_{k=0}^{\infty}\left[k(k+1)p_{k+1}(t)-k^2p_k(t)\right]\frac{a_2a_3\left(\left\langle k^2\right\rangle_t-\langle k\rangle_t\right)^3}{\left(\langle k\rangle_t\right)^2}[a_II(t)]^2\\ &= \frac{\mathrm{a_2a_3\left(\langle k\rangle_t-\left\langle k^2\right\rangle_t\right)^3}}{<\mathrm{k}>_\mathrm{t}}[\mathrm{a_II(t)}]^2.\end{aligned} \tag{50}$$

By multiplying k^2 on both sides of Eq. (49) and summing k from 0 to ∞, we have

$$\begin{aligned}\frac{d\left\langle k^2\right\rangle_t}{dt} &= \frac{d\sum\limits_{k=0}^{\infty}k^2p_k(t)}{dt}\\ &= \sum_{k=0}^{\infty}\left[k^2(k+1)p_{k+1}(t)-k^3p_k(t)\right]\frac{a_2a_3\left(\left\langle k^2\right\rangle_t-\langle k\rangle_t\right)^3}{\left(\langle k\rangle_t\right)^2}[a_II(t)]^2\\ &= \left(\langle k\rangle_t-2\left\langle k^2\right\rangle_t\right)\frac{a_2a_3\left(\left\langle k^2\right\rangle_t-\langle k\rangle_t\right)^3}{\left(\langle k\rangle_t\right)^2}[a_II(t)]^2.\end{aligned} \tag{51}$$

As a result,

$$\begin{cases}\dfrac{d\langle k\rangle_t}{dt} = \dfrac{a_2a_3\left(\langle k\rangle_t-\langle k^2\rangle_t\right)^3}{\langle k\rangle_t}[a_II(t)]^2,\\[2ex] \dfrac{d\langle k^2\rangle_t}{dt} = (\langle k\rangle_t-2\langle k^2\rangle_t)\dfrac{a_2a_3\left(\langle k^2\rangle_t-\langle k\rangle_t\right)^3}{\left(\langle k\rangle_t\right)^2}[a_II(t)]^2.\end{cases} \tag{52}$$

In Eq. (52), the second formula divided by the first formula gives

$$\frac{d\langle k^2\rangle_t}{d\langle k\rangle_t} = \frac{2\langle k^2\rangle_t}{\langle k\rangle_t} - 1. \tag{53}$$

Solving this differential equation below, we can get

$$\langle k^2\rangle_t = c\langle k\rangle_t^2 + \langle k\rangle_t, \tag{54}$$

where

$$c = \frac{G_0''(1)}{\left[G_0'(1)\right]^2} = \frac{\langle k_e\rangle}{\langle k\rangle}, \langle k_e\rangle = G_1'(1). \tag{55}$$

Substituting Eq. (55) into Eq. (52), we can solve for $\langle k\rangle_t$ as follows.

$$\langle k\rangle_t = \left(\langle k\rangle^{-4} + 4a_2a_3c^3a_I^2\int_0^t I^2(\tau)d\tau\right)^{-\frac{1}{4}}. \tag{56}$$

In this case, $\langle k^2\rangle_t$ can also be expressed by Eq. (54). And Eq. (48) and Eq. (49) can be written as

$$\begin{aligned} \frac{dp_k(t)}{dt} &= a_2a_3c^3[(k+1)p_{k+1}(t) - kp_k(t)]\langle k\rangle_t^4[a_II(t)]^2, \\ G_t'(t,x) &= a_2a_3c^3\langle k\rangle_t^4[a_II(t)]^2G_x'(t,x)(1-x), \end{aligned} \tag{57}$$

where the same $\langle k\rangle_t$ can be expressed by Eq. (56).

Next, we can construct the SEIR mean-field model with isolation time t. The definitions of $S_k(t)$, $E_k(t)$, $I_k(t)$ and $R_k(t)$ are the same as those in the previous text, where the degree distribution is $p_k(t)$. The same situation applies to $\phi_S(t)$, $\phi_E(t)$, $\phi_I(t)$ and $\phi_R(t)$. Therefore based on the degree distribution $p_k(t)$, the SEIR mean-field dynamic model of network with isolation measures is as follows:

$$\frac{dS_k(t)}{dt} = -\beta k\phi_I(t)S_k(t), \tag{58a}$$

$$\frac{dE_k(t)}{dt} = \beta k\phi_I(t)S_k(t) - \epsilon E_k(t), \tag{58b}$$

$$\frac{dI_k(t)}{dt} = \epsilon E_k(t) - \gamma I_k(t), \tag{58c}$$

$$\frac{dR_k(t)}{dt} = \gamma I_k(t), \tag{58d}$$

Solving Eq. (58a) yields

$$S_k(t) = S_k(T)\exp\left[-\beta k\int_T^t \phi_I(\tau)d\tau\right] = S_k(T)\theta^k(t), \tag{59}$$

where

$$\theta(t) = \exp\left[-\beta\int_T^t \phi_I(\tau)d\tau\right] \tag{60}$$

represents the probability that any given edge emerging from an initial susceptible node remains uninfected until time t. Because in reality, once the infected is found, they will be isolated immediately. We still assume $S_k(T) \approx 1, E_k(T) \approx 0, I_k(T) \approx 0$ and $R_k(T) \approx 0$. Furthermore, $S_k(t) = \theta^k(t)$ and

$$S(t) = \sum_{k=0}^{\infty} p_k(t)\theta^k(t) = G(t,\theta), \phi_S(t) = \frac{\sum_{k=0}^{\infty} kp_k(t)S_k(t)}{\langle k\rangle_t} = \frac{\theta G_x'(t,\theta)}{\langle k\rangle_t}. \tag{61}$$

Then, differentiating $\theta(t)$ in Eq. (60) with respect to time t leads to

$$\frac{d\theta(t)}{dt} = -\beta\theta(t)\phi_I(t). \tag{62}$$

Similarly, we get the differential equation of $\phi_I(t)$ and $\phi_E(t)$ with time t,

$$\begin{aligned}
\frac{d\phi_I(t)}{dt} &= \frac{d}{dt}\left[\frac{\sum_{k=0}^{\infty} kp_k(t)I_k(t)}{\langle k\rangle_t}\right] \\
&= \frac{\left[\sum_{k=0}^{\infty} k\frac{dp_k(t)}{dt}I_k(t) + \sum_{k=0}^{\infty} kp_k(t)\frac{dI_k(t)}{dt}\right]\langle k\rangle_t - \frac{d\langle k\rangle_t}{dt}\sum_{k=0}^{\infty} kp_k(t)I_k(t)}{(\langle k\rangle_t)^2} \\
&= \frac{\sum_{k=0}^{\infty} k\frac{dp_k(t)}{dt}I_k(t)}{\langle k\rangle_t} + \frac{\sum_{k=0}^{\infty} kp_k(t)\frac{dI_k(t)}{dt}}{\langle k\rangle_t} - \frac{\phi_I(t)}{\langle k\rangle_t}\frac{d\langle k\rangle_t}{dt} \\
&= \frac{a_2a_3\left(\left\langle k^2\right\rangle_t - \langle k\rangle_t\right)^3[a_II(t)]^2}{(\langle k\rangle_t)^2}\left\{\frac{\sum_{k=0}^{\infty} k[(k+1)p_{k+1}(t) - kp_k(t)]I_k(t)}{\langle k\rangle_t} + \phi_I(t)\right\} + \epsilon\phi_E(t) - \gamma\phi_I(t) \\
&= \frac{a_2a_3\left(\left\langle k^2\right\rangle_t - \langle k\rangle_t\right)^3[a_II(t)]^2}{(\langle k\rangle_t)^2}\left[\frac{\sum_{k=0}^{\infty} k(k+1)p_{k+1}(t) - k(k-1)p_k(t)}{\langle k\rangle_t}I_k(t)\right] + \epsilon\phi_E(t) - \gamma\phi_I(t) \\
&= \epsilon\phi_E(t) - \gamma\phi_I(t).
\end{aligned} \tag{63}$$

$$\begin{aligned}
\frac{d\phi_E(t)}{dt} &= \frac{d}{dt}\left[\frac{\sum_{k=0}^{\infty} kp_k(t)E_k(t)}{\langle k\rangle_t}\right] \\
&= \frac{\left[\sum_{k=0}^{\infty} k\frac{dp_k(t)}{dt}E_k(t) + \sum_{k=0}^{\infty} kp_k(t)\frac{dE_k(t)}{dt}\right]\langle k\rangle_t - \frac{d\langle k\rangle_t}{dt}\sum_{k=0}^{\infty} kp_k(t)E_k(t)}{(\langle k\rangle_t)^2} \\
&= \frac{\sum_{k=0}^{\infty} k\frac{dp_k(t)}{dt}E_k(t)}{\langle k\rangle_t} + \frac{\sum_{k=0}^{\infty} kp_k(t)\frac{dE_k(t)}{dt}}{\langle k\rangle_t} - \frac{\phi_E(t)}{\langle k\rangle_t}\frac{d\langle k\rangle_t}{dt} \\
&= \frac{a_2a_3\left(\left\langle k^2\right\rangle_t - \langle k\rangle_t\right)^3[a_II(t)]^2}{(\langle k\rangle_t)^2}\left\{\frac{\sum_{k=0}^{\infty} k[(k+1)p_{k+1}(t) - kp_k(t)]E_k(t)}{\langle k\rangle_t} + \phi_E(t)\right\} + \beta\phi_I(t)\frac{\theta(t)^2G_x''(t,\theta(t)) + \theta(t)G_x'(t,\theta(t))}{\langle k\rangle_t} - \epsilon\phi_E(t) \\
&= \frac{a_2a_3\left(\left\langle k^2\right\rangle_t - \langle k\rangle_t\right)^3[a_II(t)]^2}{(\langle k\rangle_t)^2}\left[\frac{\sum_{k=0}^{\infty} k(k+1)p_{k+1}(t) - k(k-1)p_k(t)}{\langle k\rangle_t}E_k(t)\right] + \beta\phi_I(t)\frac{\theta(t)^2G_x''(t,\theta(t)) + \theta(t)G_x'(t,\theta(t))}{\langle k\rangle_t} - \epsilon\phi_E(t) \\
&= \beta\phi_I(t)\frac{\theta(t)^2G_x''(t,\theta(t)) + \theta(t)G_x'(t,\theta(t))}{\langle k\rangle_t} - \epsilon\phi_E(t).
\end{aligned} \tag{64}$$

With the above derivation, the equivalent reduced-dimension model is

$$
\begin{aligned}
&\frac{d\theta(t)}{dt} = -\beta\theta(t)\phi_I(t),\\
&\frac{d\phi_E(t)}{dt} = \beta\phi_I(t)\frac{\theta^2(t)G_x''(t,\theta(t)) + \theta(t)G_x'(t,\theta(t))}{\langle k\rangle_t} - \epsilon\phi_E(t),\\
&\frac{d\phi_I(t)}{dt} = \epsilon\phi_E(t) - \gamma\phi_I(t),\\
&\frac{dI(t)}{dt} = \epsilon E(t) - \gamma I(t),\\
&\frac{dR(t)}{dt} = \gamma I(t),\\
&S(t) = G(t,\theta(t)),\\
&E(t) = 1 - S(t) - I(t) - R(t).
\end{aligned}
\tag{65}
$$

According to the definition of $\phi_R(t)$, we have

$$
\begin{aligned}
\frac{d\phi_R(t)}{dt} &= \frac{d}{dt}\left[\frac{\sum_{k=0}^{\infty} kp_k(t)R_k(t)}{\langle k\rangle_t}\right]\\
&= \frac{\left[\sum_{k=0}^{\infty} k\frac{dp_k(t)}{dt}R_k(t) + \sum_{k=0}^{\infty} kp_k(t)\frac{dR_k(t)}{dt}\right]\langle k\rangle_t - \frac{d\langle k\rangle_t}{dt}\sum_{k=0}^{\infty} kp_k(t)R_k(t)}{\left(\langle k\rangle_t\right)^2}\\
&= \frac{\sum_{k=0}^{\infty} k\frac{dp_k(t)}{dt}R_k(t)}{\langle k\rangle_t} + \frac{\sum_{k=0}^{\infty} kp_k(t)\frac{dR_k(t)}{dt}}{\langle k\rangle_t} - \frac{\phi_R(t)}{\langle k\rangle_t}\frac{d\langle k\rangle_t}{dt}\\
&= \frac{a_2a_3\left(\left\langle k^2\right\rangle_t - \langle k\rangle_t\right)^3[a_II(t)]^2}{\left(\langle k\rangle_t\right)^2}\left\{\frac{\sum_{k=0}^{\infty} k[(k+1)p_{k+1}(t) - kp_k(t)]R_k(t)}{\langle k\rangle_t} + \phi_R(t)\right\} + \gamma\phi_I(t)\\
&= \frac{a_2a_3\left(\left\langle k^2\right\rangle_t - \langle k\rangle_t\right)^3[a_II(t)]^2}{\left(\langle k\rangle_t\right)^2}\left[\frac{\sum_{k=0}^{\infty} k(k+1)p_{k+1}(t) - k(k-1)p_k(t)}{\langle k\rangle_t}R_k(t)\right] + \gamma\phi_I(t)\\
&= \gamma\phi_I(t).
\end{aligned}
\tag{66}
$$

As mentioned above from Eq. (12) to Eq. (21), the isolation model also has the following expression.

$$
\begin{aligned}
&\phi_R(t) = -\frac{\gamma}{\beta}\ln\theta(t),\\
&\phi_I(t) = \frac{(\epsilon+\gamma)}{\beta}\ln\theta(t) + \epsilon(t-T) - \frac{\epsilon}{\langle k\rangle_t}\int_T^t \theta(\tau)G_x'(\tau,\theta(\tau))d\tau + \frac{\epsilon\gamma}{\beta}\int_T^t \ln\theta(\tau)d\tau,\\
&E(t) = \beta\exp(-\epsilon t)\int_T^t \phi_I(\tau)\theta(\tau)G_x'(\tau,\theta(\tau))\exp(\epsilon\tau)d\tau,\\
&I(t) = \epsilon\exp(-\gamma t)\int_T^t \exp(\gamma\tau)E(\tau)d\tau.
\end{aligned}
\tag{67}
$$

And $R(t) = 1 - S(t) - E(t) - I(t)$, this means that $S(t)$, $E(t)$, $I(t)$, $R(t)$ can always be represented by $\theta(t)$. Among them, $E(t)$ and $I(t)$ are still canculated by the solving first-order homogeneous differential equation.

2.2.2. MCMC algorithm

Omicron BA.1 (Omicron strains data in UK from PHE), a variant of COVID-19, was first detected in South Africa on November 9, 2021. As of December 2, 2021, it has affected at least 23 countries and regions around the world, including developed countries such as the United Kingdom, Japan and Netherlands. Among them, the British Department of Public Health disclosed the number of new cases in the initial stage of infection of this strain (Public Health England). Therefore,

according to the literature (Haario et al., 2006; Gamerman & Lopes, 2006), MCMC (Markov Chain Monte Carlo) Metropolis-Hastings (M-H) algorithm could be used to estimate the relevant parameters of the transmission of this strain in the UK.

By solving the model (1) with MATLAB, the relationship between the numerical solution of the equation and the new cases data is as follows:

$$y(t) = y^*(t) + \epsilon, t = 0, 1, \ldots, 19, \tag{68}$$

where

$$y^*(t) = \sum_{k=0}^{K_{max}} I_k(t) \tag{69}$$

with $y(0) = y^*(0)$ and ε is assumed to follow the standard normal distribution $N(0, \sigma^2)(\sigma = 0.1)$. If we denote the parameter vector by $\Theta = (\beta, \epsilon, \gamma)$, the maximum likelihood function for the data $Y = \{y(t)\}_0^{T_0=19}$ is

$$L(Y|\Theta) = \prod_{t=0}^{T_0} \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{ -\frac{[y(t) - y^*(t)]^2}{\sigma^2} \right\}. \tag{70}$$

Then, the joint posterior distribution of the parameters is $P(\Theta|Y) \propto L(Y|\Theta)P(\Theta)$. For many biological models, one may only know the fact that the parameters of model are greater than zero, so non-information prior distribution is adopted (Gamerman & Lopes, 2006; Xiao et al., 2012). They generally assume $P(\Theta) \propto$ constant, so the joint posterior distribution of the parameters becomes

$$P(\Theta|Y) \propto L(Y|\Theta). \tag{71}$$

In order to estimate parameters in model (1), at each iteration, given the initial value $\Theta(0)$ we update the sample $\Theta*$ by random walk (Gamerman & Lopes, 2006) with the maximum walk values $\omega = (1 \times 10^{-4}, 1 \times 10^{-4}, 1 \times 10^{-4})$. Then, with respect to the new parameters $\Theta*$, we calculate the new cases of model (1) based on Eq. (69) to get the posterior likelihood $P(\Theta*|Y)$ based on Eqs. (70) and (71). The new values $\Theta*$ of parameters will be accepted with probability

$$\min\left(1, \frac{P(\Theta^*|Y)}{P(\Theta|Y)}\right). \tag{72}$$

3. Results

3.1. Parameter estimation on scale-free networks

Firstly, in recent years, people have conducted a lot of research on the community structure of complex networks such as social networks, biological networks and technological networks (Boccaletti et al., 2006; Milo et al., 2004; Song et al., 2005; Szabaó & Fath, 2007; Szolnoki et al., 2008; Szolnoki & Perc, 2009). It is found that these networks often have dynamic structural characteristics. For example, in a social network, the contacts and connections between people are constantly changing. With the flow of population, people constantly establish new connections and lose old ones (Brockmann et al., 2006; Gonzalez et al., 2008). Brockmann et al. (Brockmann et al., 2006) published an empirical research paper on people's mobility behavior in Nature. Through tracking the consumption records of 464.670 bank checks in the United States, it is found that the distance distribution of human travel meets the power law distribution, indicating that human flow behavior is similar to the "Levy flights" behavior of animal migration (Klafter et al., 1996; Viswanathan et al., 1996).

Secondly, Joel Mossong et al. (Mossong et al., 2008; Prem et al., 2017) conducted a population-based prospective mixed model survey in eight European countries, which recorded the number of contacts between participants and different individuals in a day, among which the average number of contacts reported in the United Kingdom was 11.7. In this paper, each person's daily contact with others was only calculated once. Therefore, we consider the British Omicron propagation model on a scale-free network (power-law distribution) with an average degree of 11.7.

3.1.1. Estimation and fitting on scale-free networks

Analysis was performed on a scale-free network with average degree of $\langle k \rangle = 11.7$ (index $\alpha = 2.77$) and $\Theta(0) = (0.04, 0.5, 0.143)$, which is obtained from (Chen et al., 2023; Zhang et al., 2022). For data reliability, we select data between November 27, 2021 and December 16, 2021 (blue circles in Fig. 2). According to these actual data, the number of people in each state for the UK on November 27, 2021 is $S(0) = 68414989$, $E(0) = 6$, $I(0) = 2$, $R(0) = 0$. Based on the above initial conditions of parameters of model (1) and degree distributions in Table 1, MCMC algorithm is run for 10,000 iterations with a burn-in of 3000 iterations

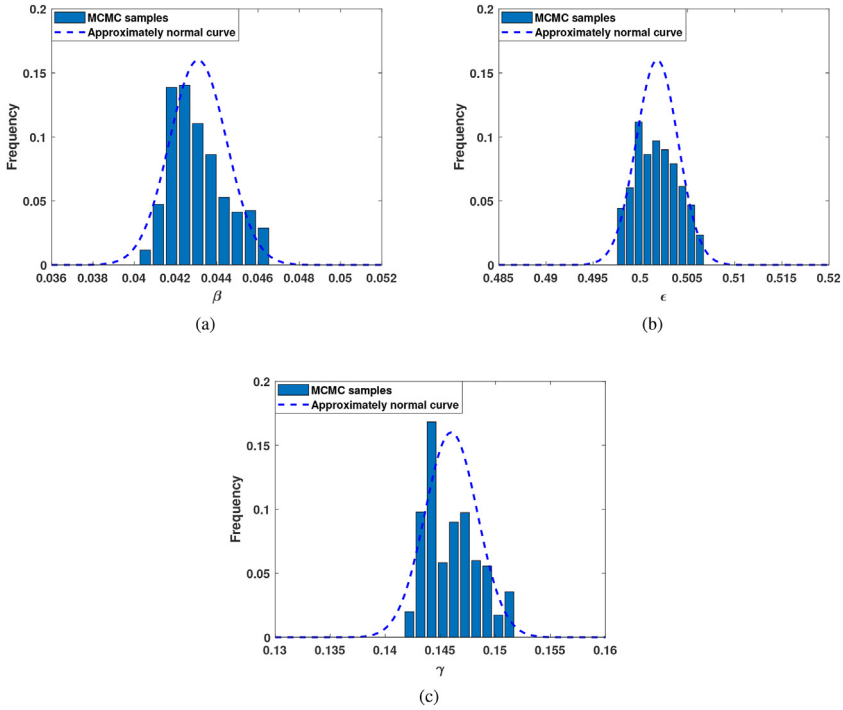


Fig. 1. MCMC is used to estimate the value and range of parameters β, ϵ, γ of model (1). (a), (b) and (c) corresponding empirical distributions and approximate normal curves of parameters β, ϵ and γ are given respectively. Among them, $\beta = 0.0431$(95 %CI : [0.04307, 0.04313]),$\epsilon = 0.5018$(95 %CI : [0.50176, 0.50184]),$\gamma = 0.1460$(95 %CI : [0.14595, 0.14605]).

to estimate the parameters β, ϵ, γ, and Fig. 1 show their empirical distribution and the curves of the approximatively normal distributions.

Based on the scale-free network with an average degree of 11.7 and the parameter values fitted by the above MCMC algorithm, the fitting plot of the Omicron strain data reported in the UK from November 27, 2021 to December 16, 2021 can be given in model (1).

The solid red line in Fig. 2(a) illustrates the median of the 7000 simulations, and the shaded portion represents the 95 % confidence interval. Fig. 2(b) shows the relative errors $(y - y^*)/y^*$ between the theoretical values and reported data. All errors are distributed on both sides of the zero horizontal line, indicating a good fit between the model and the data. November 28 and December 2 saw a surge in reported cases, primarily due to the initial underreporting of Omicron variant infections when testing was not yet widely available. As testing efforts were ramped up later, the case numbers on those dates reflected a sudden increase, capturing previously undetected and unconfirmed infections. According to the parameters obtained by fitting and the network with average degree 11.7, the basic reproductive number of Omicron in UK can be calculated as $R_0 = \frac{\beta}{\gamma}\frac{\langle k^2\rangle}{\langle k\rangle} = \frac{0.0431}{0.1460} \times \frac{203.5067}{11.7} \approx 5.1347(95\%\text{CI} : [5.1294, 5.1401])$ according to Eq. (29).

Then based on the fitting results of the parameters on the scale-free network, the change of degree distribution and Omicron propagation under tracing isolation are analyzed. The United Kingdom and other European countries did not adopt tracing policy during the transmission of Omicron. It is interesting to consider the changes in the degree distribution of the scale-free network with average degree 11.7 and the changes in the values associated with the transmission of the Omicron if the UK adopted tracing isolation.

Table 1 in Appendix C shows the degree distribution of a scale-free network with average degree 11.7. In this table, the degree distribution has no isolation measures. In other words, this is the initial degree distribution.

We assume that the tracing isolation is implemented on November 28. That is, the daily positive cases, including the close contacts and indirect contacts of positive cases are isolated, which will lead to changes in the degree distribution from November 29. Similarly, if we consider the implementation of the tracing isolation at different time on November 29, December 1, December 6, December 11, and December 16, then the changes in the degree distribution are also different. Besides, if isolation is carried out at different time, the number of the susceptible, exposed, infected and recovered will also vary.

These results can be obtained by MATLAB and parameters a_I, a_E, a_1, a_2, a_3 can be obtained from relevant literature. The infected can only be found through nucleic acid detection. According to the literature (Kanji et al., 2023; Wen Z et al., 2020), the most commonly used RT-PCR method has a false negative probability 0.1, so it is recommended to set $a_I = 0.9$. Due to low viral load and high CT value, the exposed cannot be detected by nucleic acid detection (Chen et al., 2023), so $a_E = 0$ is

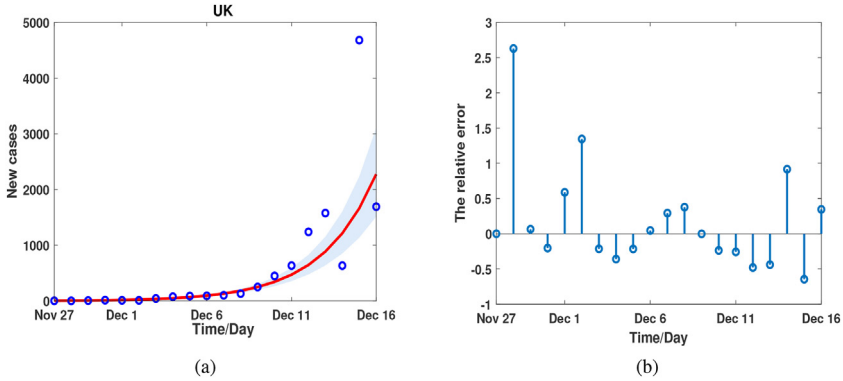


Fig. 2. Model (1) fits the new outbreak data from November 27, 2021 to December 16, 2021 in Omicron, United Kingdom. (a) The blue circles show the number of the reported new cases, the shaded region shows 95 % range of 7000 simulations, and the solid red curve is the median values of the 7000 simulations. (b) The relative errors between the theoretical value of model (1) and the real data.

recommended. Li Qian, Xiao Yanni et al. (Li et al., 2020) discussed the maximum isolation rate that can be achieved under the tracing isolation, so $a_1 = a_2 = a_3 = 0.9$ is selected. Based on these parameters, the correlation analysis of degree distribution is carried out. For tracing isolation starting on November 28, the degree distribution for November 29 can be derived. And Appendix C includes an example of the degree distribution on November 29 in Table 2. We can also figure out how the degree distribution has changed since December 1. No more examples here.

Obviously, the accuracy of the degree distribution is relatively large. When the degree distribution at different time are drawn on the same graph, they will coincide as a line, which cannot be observed well by the naked eye. Consequently, the difference between the degree distribution at each moment and the initial degree distribution for tracing isolation starting on November 28 is shown in Fig. 3(a) below. The degree distribution at different time changes outward from the horizontal axis, that is, the line closest to the horizontal axis is the difference between the first day of isolation and the initial degree distribution. The next line below is the difference between the degree distribution on the second day of isolation and the initial degree distribution, and so on. The top line is the difference between the degree distribution on the last day of isolation and the initial degree distribution. Similarly, we also make a log-log scale plot of the degree distribution and show it in Fig. 3(b) below. In addition, for the degree distribution, under the tracing isolation, the changes in the number of the susceptible, exposed, infected and recovered in society are shown in Fig. 3(c) below.

For tracing isolation starting on November 29, December 1, December 6, December 11 and December 16, there are the degree distribution images and the images of the number of the susceptible, exposed, infected and recovered in society.

During the simulation, it is found that if the tracing isolation policy is implemented on November 28, November 29 and December 1, then all people would be isolated on the 38th day of the disease spread. Similarly, if the tracing isolation policy is implemented on December 6, December 11, and December 16, all people will be isolated on the 37th, 36th and 35th day of the disease transmission, respectively. Obviously, the later the isolation time, the shorter the transmission duration of the strain in society. That is, the later the isolation time, the earlier all people will be isolated. This is because the later the isolation time, the greater the number of the infected until the implementation of the tracing isolation policy, then the more people need to be isolated. In this way, all people will be isolated very quickly. Therefore, the degree distribution images and the change images of the number of people in each state are given on the basis of the above days.

Figs. 3(a), 4(a), 5(a), 6(a), 7(a) and 8(a) describe the changes in the difference between each time degree distribution and the initial degree distribution after the implementation of tracing isolation measures. Figs. 3(b), 4(b), 5(b), 6(b), 7(b) and 8(b)

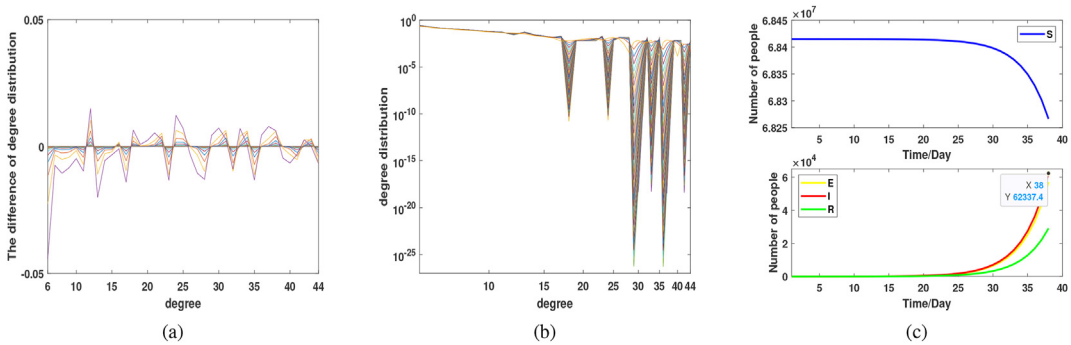
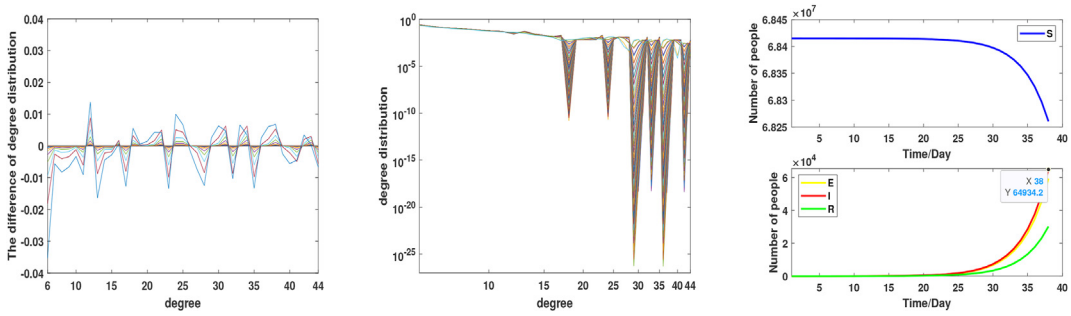
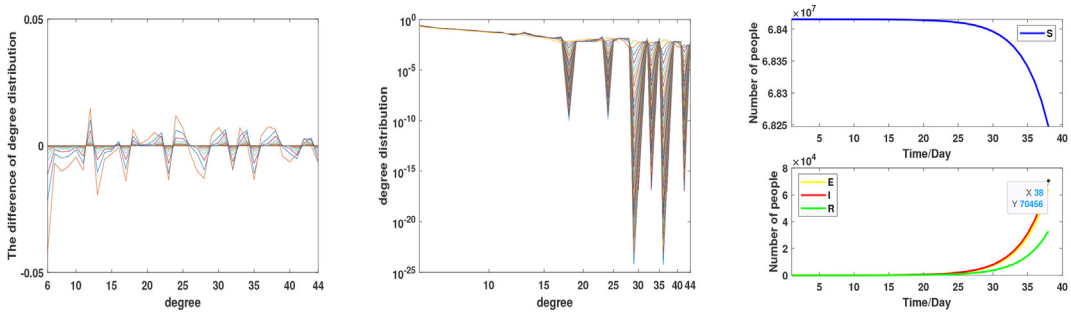


Fig. 3. The degree distribution images that began to be isolated on November 28 and the changes in the number of people in each state.



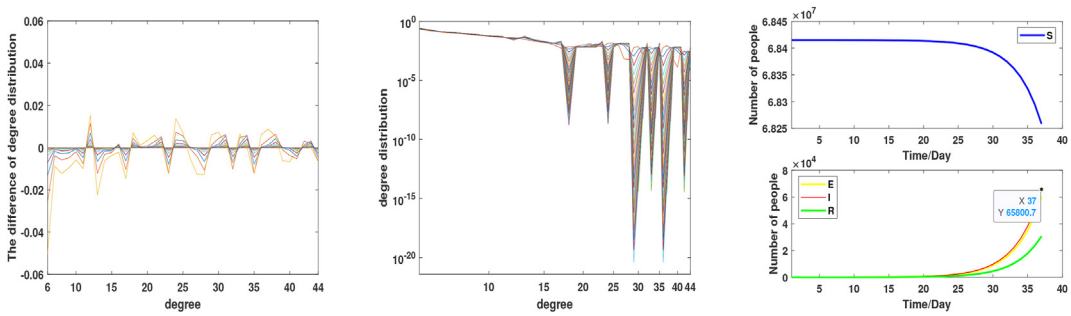
(a) (b) (c)

Fig. 4. The degree distribution images that began to be isolated on November 29 and the changes in the number of people in each state.



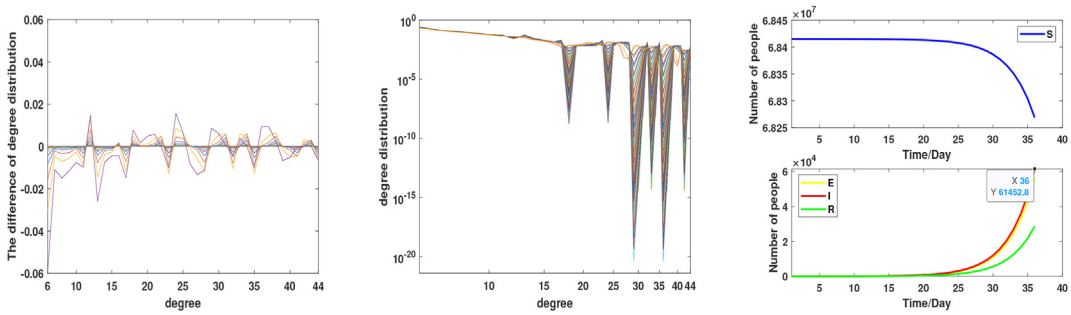
(a) (b) (c)

Fig. 5. The degree distribution images that began to be isolated on December 1 and the changes in the number of people in each state.



(a) (b) (c)

Fig. 6. The degree distribution images that began to be isolated on December 6 and the changes in the number of people in each state.



(a) (b) (c)

Fig. 7. The degree distribution images that began to be isolated on December 11 and the changes in the number of people in each state.

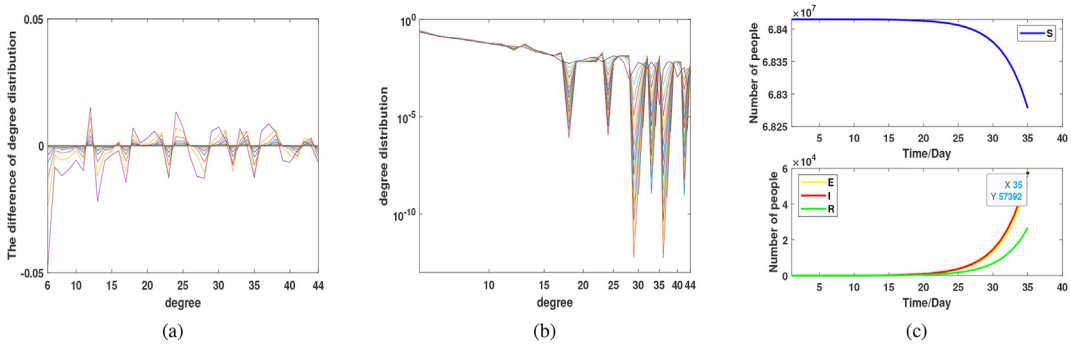


Fig. 8. The degree distribution images that began to be isolated on December 16 and the changes in the number of people in each state.

exhibit a log-log scale plot of the degree distribution. These figures show the changes in degree distribution at different times after the implementation of isolation measures. At the initial moment, the network follows a power-law distribution. However, as the isolation policy is gradually implemented, the connectivity in the network is significantly affected. It can be seen that nodes with degree less than 15 still show the characteristics of a power-law distribution, indicating that the low degree area of the network is relatively stable and has not been seriously disturbed. However, the part with degree greater than 15 shows obvious noise, which may be due to the isolation of high degree nodes, resulting in the fragmentation or instability of the network structure. Figs. 3(c), 4(c), 5(c), 6(c), 7(c) and 8(c) show the changes in the number of people in each state of disease transmission before all people are isolated. In the early stages of an infectious disease, the later the isolation measures are implemented, the higher the number of infected individuals on the last day in society. This is because if isolation is delayed, the initial infected individuals are not promptly isolated, allowing more latent and infected carriers to interact with the people in society, which accelerates the spread and expansion of the virus. As the transmission chain extends, the number of infected individuals in society increases significantly by the final isolation day. Besides, after the disease has spread for a period of time (that is, the tracing isolation policy is implemented at December 6, December 11, and December 16), the later the isolation measures are implemented, the smaller the number of infected individuals on the last day in society. This is because once the large numbers of infected people, their close contacts and their indirect contacts are isolated, it effectively reduces the vast majority of sources of infection. Furthermore, the later the isolation, the fewer the number of the infected in society.

3.2. Changes in the first moment and the second moment

According to the previous Eq. (52), the changes of the first moment and the second moment of the degree distribution with time under different isolation time can be obtained.

For the tracing isolation on November 28, November 29, December 1, December 6, December 11 and December 16, the changes of the first moment and the second moment of the degree distribution over time are as follows.

In Fig. (9), the red, green, blue, black, pink and brown lines represent the isolation from November 28, November 29, December 1, December 6, December 11 and December 16, respectively.

As can be seen from the figure, the first moment and the second moment of degree distribution become smaller and smaller with the later the isolation time. This is because at the beginning of the transmission of this strain, the later the

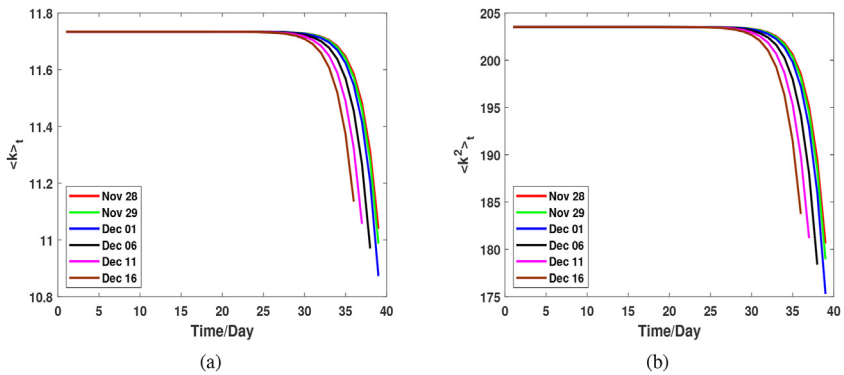


Fig. 9. The changes in the first moment and the second moment are isolated at different tracing isolation time.

isolation time, the more nodes are isolated. Then the total number of nodes in the network in society has been decreasing, so the first moment and the second moment will be smaller and smaller. That is, the later the isolation, the more people are isolated and the average number of contacts per person per day will be less.

4. Discussion and conclusion

Due to the complexity of network structure, it is a hot issue for policy makers to develop network-based measures for infectious disease control and intervention (Pellis et al., 2015). Since the end of 2019, COVID-19 has brought great harm to human life and health and also brought huge losses to the economies of countries around the world. In order to scientifically study the law of transmission and prevention and control measures of COVID-19 in the new situation, it is necessary to establish a network-based dynamic model of COVID-19.

This paper is in the process of deducing the change of the degree distribution under the tracing isolation (i.e. the isolation of the infected, the close contacts, and the indirect contacts). Based on the definition of probability generating function and relevant knowledge of excess degree distribution, the average number of d-order neighbors of a node on the network can be canculated. Then assuming that the isolation starts at a certain time, we can get the total density of the isolated nodes in the time period t. It is analyzed how many nodes with degree k decrease because of isolation and how many nodes with degree greater than k become k because of isolation. After ignoring the higher order term, we derive the differential equations of degree distribution related to time t and the coupling equation of the first and the second moment only related to the number of the infected. Hence we acquire the relationship between the first and the second moment.

Based on the data of the outbreak of the Omicron in the United Kingdom at the end of 2021, three relevant parameters of Omicron transmission on the scale-free network were estimated by MCMC algorithm. According to the above differential equation about time t of dynamic network degree distribution and the SEIR mean-field dynamics model with isolation, the change of degree distribution and the number of the susceptible, exposed, infected and recovered under different isolation time are shown by using the obtained parameter values. Tracing isolation at different time will result in different duration of the disease to spread in society. For the degree distribution of the network, we describe the changes in the difference between each time distribution and the initial time distribution after the implementation of tracing isolation measures. In the process of analyzing the images in the main text above, the changes in the number of people in each state of disease transmission can be obtained.

The method of deriving the average d-order neighbor number of a node in this paper can be applied to the study of the spread of passive worms in P2P networks (Zhang et al., 2018). The method of dimensionality reduction which gets the exact expression of the final size and the solution of $S(t)$, $E(t)$, $I(t)$, $R(t)$ of the SEIR mean-field dynamics model is also suitable for more complex mean-field dynamics models. In the process of deriving degree distribution and other differential equations related to time t, the change of degree can be inferred by analyzing the broken edges of nodes. And the problems of link reconstruction in network can use this method. SEIR mean-field dynamics model in network including isolation policy can be utilized as an important model framework to study the development trend of COVID-19 in other countries or other domestic provinces. We can estimate the parameter by MCMC algorithm and analyze the number of the infected, which can be applied in the study of other strains.

The research methods and results in this paper provide guidance for the dynamic analysis of COVID-19 transmission network under the tracing isolation. The disadvantage is that clustering is not involved when considering the network. In addition, the number of the infected is more accurate and reasonable if the delays that may arise in the process of isolation are included in the model, which can be left for further study.

CRediT authorship contribution statement

Jiangmin Li: Writing – original draft. **Zhen Jin:** Writing – review & editing. **Ming Tang:** Writing – review & editing.

Statements and declarations

The authors declare that they have no conflict of interest.

Data availability

https://ukhsa-dashboard.data.gov.uk/respiratory-viruses/covid-19#cases.

Funding

This work was supported by the National Natural Science Foundation of China (Grant No.12231012, No.U23A20331) and Key Research and Development Project in Shanxi Province, China (Grant No.202003D31011/GZ).

Declaration of interest statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We also thank the reviewers and editors for their comments and suggestions. This work was supported by the National Natural Science Foundation of China (Grant No.12231012, No.U23A20331) and Key Research and Development Project in Shanxi Province, China (Grant No.202003D31011/GZ).

Appendix A. The derivation of the average number of neighbors

The number of edges attached to a node other than the edge we arrived along is called the excess degree of the node and it is just one less than the total degree. Since the node at the end of an edge always has degree at least 1 (because of that edge) the minimum value of the excess degree is zero. Excess degree distribution is the probability distribution, for a node reached by following an edge, of the number of other edges attached to that node (Newman, 2010). In other words, the excess degree distribution is the distribution of the number of edges that go out as the edge enters a node. Thus the probability that we reach a node of degree k upon following an edge is $\frac{kp(k)}{\langle k\rangle}$, that is, the probability of connecting to a node with excess degree k is

$$\frac{(k+1)p(k+1)}{\langle k\rangle} \triangleq q(k). \tag{73}$$

Then, the generating function corresponding to the excess degree distribution is

$$G_1(x) = \sum_{k=0}^{\infty} q(k)x^k \tag{74}$$

and it can be concluded that

$$G_1'(x) = \sum_{k=0}^{\infty} kq(k)x^{k-1}, \tag{75}$$

$$G_1'(1) = \sum_{k=0}^{\infty} kq(k) = \sum_{k=0}^{\infty} k\frac{(k+1)p(k+1)}{\langle k\rangle} = \sum_{k=1}^{\infty} \frac{k(k-1)p(k)}{\langle k\rangle} = \frac{\langle k^2\rangle - \langle k\rangle}{\langle k\rangle}. \tag{76}$$

In this paper, we are particularly concerned about the average number of d-order neighbors of a node. Based on the knowledge of the network, it is clear that the number of first-order neighbors of a node is the degree of the node. Then in the network, any node can be selected to obtain the average number of the first-order neighbors of the node as the average degree of the network $C_1 \triangleq \langle k\rangle$.

Next, the excess degree distribution is used to derive the average number of second-order neighbors of a node). Firstly, the probability $p_k^{(2)}$ that a node has k second-order neighbors is given by $p_k^{(2)} = \sum_{m=0}^{\infty} p(m)p^{(2)}(k|m)$, where $p^{(2)}(k|m)$ is the probability of having k second-order neighbors given that there are m first-order neighbors and $p(m)$ is the degree distribution of the network. We can easily see that the number of second-order neighbors of a node is equal to the sum of the excess degrees of each of the first-order neighbors. Suppose the excess degrees of the m first-order neighbors are $j_1, j_2, j_3, \ldots, j_m$, respectively. The probability distribution of the sum of these m integers has the generating function $[G_1(x)]^m$. That is, $\sum_{k=0}^{\infty} p^{(2)}(k|m)x^k = [G_1(x)]^m$. Therefore, we get

$$G^{(2)}(x) = \sum_{k=0}^{\infty} p_k^{(2)} x^k = \sum_{k=0}^{\infty}\left[\sum_{m=0}^{\infty} p(m)p^{(2)}(k|m)\right]x^k = \sum_{m=0}^{\infty} p(m)\sum_{k=0}^{\infty} p^{(2)}(k|m)x^k = \sum_{m=0}^{\infty} p(m)[G_1(x)]^m = G_0(G_1(x)). \tag{77}$$

Differentiating the above equation and letting $x = 1$, we get the average number C_2 of second-order neighbors of a node, that is

$$C_2 \triangleq G_0'(1)G_1'(1) = \langle k^2\rangle - \langle k\rangle, \tag{78}$$

where $G_0'(1) = \langle k\rangle$ and $G_1'(1) = \frac{\langle k^2\rangle - \langle k\rangle}{\langle k\rangle}$.

Similarly, if there are m $d-1$-order neighbors, then the generating function for the number of d-order neighbors can be

represented by $G^{(d)}(x) = \sum_{k=0}^{\infty} p_k^{(d)} x^k = \sum_{k=0}^{\infty}\sum_{m=0}^{\infty} p_m^{(d-1)} p^{(d)}(k|m) x^k = G^{(d-1)}(G_1(x)) = G_0(G_1(\ldots(G_1(x))))$.

Similarly, the average number C_d of d-order neighbors is given by

$$C_d = \left(\frac{C_2}{C_1}\right)^{d-1} \cdot C_1. \tag{79}$$

Appendix B. The change of degree distribution after isolation

Obviously, we know that the global density of infected nodes at time T is

$$I(T) = \sum_{k=0}^{\infty} p(k) I_k(T). \tag{80}$$

Suppose that the probability of infected nodes being found in the network is a_I and the probability of exposed nodes being found is a_E. Since the infected nodes, close contact nodes and indirect contact nodes are isolated at time T, then the density of the isolated infected nodes at time T is $a_I I(T)$. Therefore we have the density of the isolated close contact nodes at time T is $a_I a_1 C_1(T) I(T)$ and the density of the isolated indirect contact nodes is

$$a_I a_2 C_2(T) I(T) \triangleq M(T), \tag{81}$$

where $C_1(T) = \langle k \rangle, C_2(T) = \langle k^2 \rangle - \langle k \rangle, C_d(T) = \left[\frac{C_2(T)}{C_1(T)}\right]^{d-1} C_1(T)$, the density of isolated third-order neighbor nodes at time T is

$$a_I a_3 C_3(T) I(T) \triangleq L(T). \tag{82}$$

In a word, the density of isolated nodes in the whole network at time T is

$$a_I I(T) + a_I a_1 C_1(T) I(T) + a_I a_2 C_2(T) I(T) = a_I I(T)(1 + a_1 C_1(T) + a_2 C_2(T)) \triangleq Q(T). \tag{83}$$

So the total density of isolated nodes in the entire network in the Δt time period is $Q(T)\Delta t$.

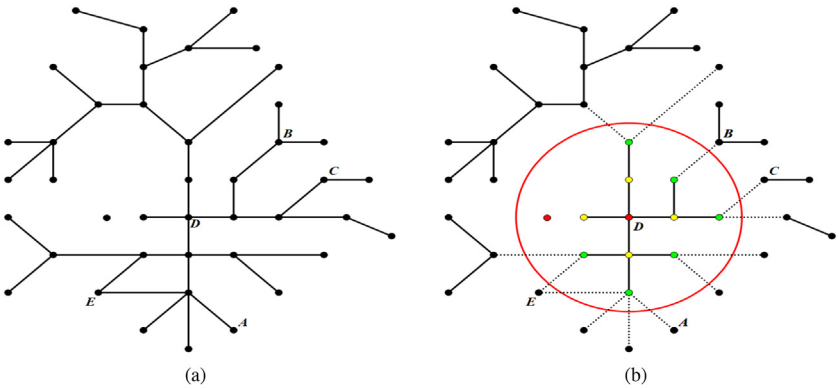


Fig. 10. This model diagram simply shows how the degrees of nodes change during the evolutionary process of tracing isolation. Red nodes represent the infected nodes, yellow nodes represent close contact nodes and green nodes represent indirect contact nodes.

In Fig. 10(a) and (b), point A is a node with degree 1 at time T and its degree will become 0 because the only edge is cut off after isolation; point B is a node with degree 3 at time T and its degree will become 2 because one edge is cut off after isolation; point C is a node with degree 2 at time T and its degree will become 1 because one edge is cut off after isolation; point D is a node with degree 4 at time T and its degree will become 0 due to isolation; point E is a node with degree 2 at time T and its degree will become 0 because two edges are cut off after isolation. There are also cases where nodes with degree 0 are isolated. Based on this analysis, we can discuss the changes in the total number of nodes with different degrees.

The change of the total number of nodes with degree $k(k > 0)$ at time T of the remaining network after tracing isolation can be divided into the following situations:

(1) Isolation: isolated as infected nodes, close contact nodes, indirect contact nodes. These nodes leave the network directly after being isolated.
(2) In the process of isolating the indirect contact nodes, some nodes may change their degree from k to $k-1$, $k-2$, ..., 0 because of the edge connection with one or more indirect contact nodes. It can be classified and discussed according to the size relationship between the degree k and the number of isolated indirect contact nodes. Since $M(T)$ discussed

above is the density of the indirect contact nodes, we can use $D(T)$ to represent the number of the indirect contact nodes isolated at time T.

If $k > D(T)$, then there are some nodes whose degree changes from k to $k-1$ or some nodes whose degree changes from k to $k-2$. This situation may extend to the degree of the node from k to $k-D(T)$. If $k \leq D(T)$, then there are some nodes whose degree changes from k to $k-1$ or some nodes whose degree changes from k to $k-2$. This situation may extend to the degree of the node from k to 0. The node density with degree k isolated at time T is p(k)Q(T). In degree-irrelevant network, the probability of starting from a node at time T (assuming that the degree of the node is h) along an edge to reach a first-order neighbor node with degree m is $p(m|h) = \frac{mp(m)}{\langle k \rangle}$.

Then the probability of starting from the first-order neighbor node with degree m to reach a second-order neighbor node with degree l is $p(l|m) = \frac{lp(l)}{\langle k \rangle}$. In the same way, the probability of starting from the second-order neighbor node with degree l to reach a third-order neighbor node with degree k is

$$p(k|l) = \frac{kp(k)}{\langle k \rangle}. \tag{84}$$

Therefore the probability of arriving at a third-order neighbor node with degree k from a node is

$$\sum_{m,l=0}^{\infty} p(m|h)p(l|m)p(k|l) = \sum_{m,l=0}^{\infty} \frac{mp(m)}{\langle k \rangle} \frac{lp(l)}{\langle k \rangle} \frac{kp(k)}{\langle k \rangle} = \frac{kp(k)}{\langle k \rangle}. \tag{85}$$

That is, in the degree-irrelevant network, the probability of starting from a node to a third-order neighbor node with degree k is the probability that a second-order neighbor node of the node connects to a third-order neighbor node with degree k. Because what we consider at the beginning is the node density, the probability that the node is an isolated indirect contact node at time T is $M(T)$.

So the probability that an indirect contact node connects to a third-order neighbor with degree k is defined as

$$M(T)\frac{kp(k)}{\langle k \rangle} \triangleq q_k(T). \tag{86}$$

Then in this degree-irrelevant network, the probability of two indirect contact nodes connecting to the same third-order neighbor with degree k is $q_k^2(T)$. We can continue to do so, in this degree-irrelevant network the probability of $n(n \leq k)$ indirect contact nodes connecting to the same third-order neighbor with degree k is $q_k^n(T)$.

If $k > D(T)$, we have the node density from degree k to degree $k-1$ is $q_k(T)L(T)$, from degree k to degree $k-2$ is $q_k^2(T)L(T)$, so we have the node density from degree k to degree $k-D(T)$ is $q_k^{D(T)}(T)L(T)$ at time T. As a result, the total density of this type nodes at time T is

$$q_k(T)L(T) + q_k^2(T)L(T) + \cdots + q_k^{D(T)}(T)L(T) = q_k(T)L(T)\cdot\frac{1-q_k^{D(T)}(T)}{1-q_k(T)}. \tag{87}$$

If $k \leq D(T)$, we have the node density from degree k to degree $k-1$ is $q_k(T)L(T)$ and from degree k to degree $k-2$ is $q_k^2(T)L(T)$, so we have the node density from degree k to degree 0 is $q_k^k(T)L(T)$ at time T. As a result, the total density of this type nodes at time T is

$$q_k(T)L(T) + q_k^2(T)L(T) + \cdots + q_k^k(T)L(T) = q_k(T)L(T)\cdot\frac{1-q_k^k(T)}{1-q_k(T)}. \tag{88}$$

In fact, $p_k(t)$ represents the total density of nodes with degree k at time t and the nodes satisfy the normalization condition: $S(t)+E(t)+I(t)+R(t)=1$. According to the above derivation of the degree density change of each node after tracing isolation, we can easily obtain the remaining network degree distribution $p_k(T+\Delta t)$ at the tracing isolation time $T+\Delta t$ as follows:

(1) In the remaining network, the density change of nodes with degree 0 at time $T+\Delta t$ is analyzed as follows:

Removal situation: At time T, the nodes with degree 0 in the network are isolated as infected nodes with density $P(0)Q(T)$.

Entry situation: At time T, the degree of the node in the network changes from $m(m \leq D(T))$ to 0 due to the isolation of the indirect contact nodes. And these nodes' density is $\sum_{m=1}^{D(T)} q_m^m(T)L(T)$.

As a result, the node density with degree 0 at time $T+\Delta t$ is

$$p(0) - p(0)Q(T)\Delta t + \sum_{m=1}^{D(T)} q_m^m(T)L(T)\Delta t, \tag{89}$$

then

$$p_0(T+\Delta t) = \frac{p(0) - p(0)Q(T)\Delta t + \sum_{m=1}^{D(T)} q_m^m(T)L(T)\Delta t}{1 - Q(T)\Delta t}. \tag{90}$$

(2) In the remaining network, when $0 < k \leq D(T)$, the change of node density with degree k at time $T + \Delta t$ is as follows:

Removal situation: At time T, the nodes with degree k in the network are isolated as infected nodes or close contact nodes or indirect contact nodes and their density is $p(k)Q(T)$; at time T, the degree of the node changes from k to $k-1, k-2, \ldots, 0$ due to the isolation of the indirect contact nodes, and these nodes' density is

$$q_k(T)L(T) + q_k^2(T)L(T) + \cdots + q_k^k(T)L(T) = q_k(T)L(T)\frac{1 - q_k^k(T)}{1 - q_k(T)}. \tag{91}$$

Entry situation: At time T, the degree of the node changes from $m(m > k)$ to k due to the isolation of the indirect contact nodes, but the degree of a node whose degree is larger than $D(T) + k$ can not be reduced to k due to the isolation of the indirect contact nodes. Therefore the range of m can be limited to $k < m \leq D(T) + k$ and these nodes' density is $\sum_{m=k+1}^{k+D(T)} q_m^{m-k}(T)L(T)$.

As a result, the node density with degree $k(0 < k \leq D(T))$ at time $T + \Delta t$ is

$$p(k) - p(k)Q(T)\Delta t - q_k(T)L(T)\frac{1 - q_k^k(T)}{1 - q_k(T)}\Delta t + \sum_{m=k+1}^{k+D(T)} q_m^{m-k}(T)L(T)\Delta t, \tag{92}$$

then

$$p_k(T+\Delta t) = \frac{p(k) - p(k)Q(T)\Delta t - q_k(T)L(T)\frac{1-q_k^k(T)}{1-q_k(T)}\Delta t + \sum_{m=k+1}^{k+D(T)} q_m^{m-k}(T)L(T)\Delta t}{1 - Q(T)\Delta t}. \tag{93}$$

(3) In the remaining network, when $k > D(T)$ the change of node density with degree k at time $T + \Delta t$ is as follows:

Removal situation: At time T, the nodes with degree k in the network are isolated as infected nodes or close contact nodes or indirect contact nodes and their density is $p(k)Q(T)$; at time T, the degree of the node changes from k to $k-1, k-2, \ldots, k-D(T)$ due to the isolation of the indirect contact nodes, and these nodes' density is

$$q_k(T)L(T) + q_k^2(T)L(T) + \cdots + q_k^{D(T)}(T)L(T) = q_k(T)L(T)\frac{1 - q_k^{D(T)}(T)}{1 - q_k(T)}. \tag{94}$$

Entry situation: At time T, the degree of the node changes from $m(m > k)$ to k due to the isolation of the indirect contact nodes, but the degree of a node whose degree is larger than $D(T) + k$ can not be reduced to k due to the isolation of the indirect contact nodes. Therefore the range of m can be limited to $k < m \leq D(T) + k$ and these nodes' density is $\sum_{m=k+1}^{k+D(T)} q_m^{m-k}(T)L(T)$.

As a result, the node density with degree $k(k > D(T))$ at time $T + \Delta t$ is

$$p(k) - p(k)Q(T)\Delta t - q_k(T)L(T)\frac{1 - q_k^{D(T)}(T)}{1 - q_k(T)}\Delta t + \sum_{m=k+1}^{k+D(T)} q_m^{m-k}(T)L(T)\Delta t, \tag{95}$$

then

$$p_k(T+\Delta t) = \frac{p(k) - p(k)Q(T)\Delta t - q_k(T)L(T)\frac{1-q_k^{D(T)}(T)}{1-q_k(T)}\Delta t + \sum_{m=k+1}^{k+D(T)} q_m^{m-k}(T)L(T)\Delta t}{1 - Q(T)\Delta t}. \tag{96}$$

Because in the early stage of the spread of infectious diseases, the probability of two or more indirect contact nodes connecting to the same third-order neighbor node with degree k is very low. That is to say, some nodes have a low probability

of degree changing from k to $k-2$, …, due to the isolation of indirect contact nodes. And this situation is no longer considered in the model. Then we can estimate the formula related to $M(T)\frac{kp(k)}{\langle k\rangle} \triangleq q_k(T)$:

$$\begin{aligned}
&\sum_{m=1}^{D(T)} q_m^m(T)L(T) \approx q_1(T)L(T),\\
&\sum_{m=k+1}^{k+D(T)} q_m^{m-k}(T)L(T) \approx q_{k+1}(T)L(T),\\
&q_k(T)L(T) + q_k^2(T)L(T) + \cdots + q_k^k(T)L(T) \approx q_k(T)L(T),\\
&q_k(T)L(T) + q_k^2(T)L(T) + \cdots + q_k^{D(T)}(T)L(T) \approx q_k(T)L(T).
\end{aligned} \tag{97}$$

Then according to the above estimation, above formulas can be estimated as follows:

$$\begin{aligned}
&k=0, \quad p_0(T+\Delta t) \approx \frac{p(0) - p(0)Q(T)\Delta t + q_1(T)L(T)\Delta t}{1-Q(T)\Delta t},\\
&0<k\le D(T), \quad p_k(T+\Delta t) \approx \frac{p(k) - p(k)Q(T)\Delta t - q_k(T)L(T)\Delta t + q_{k+1}(T)L(T)\Delta t}{1-Q(T)\Delta t},\\
&k>D(T), \quad p_k(T+\Delta t) \approx \frac{p(k) - p(k)Q(T)\Delta t - q_k(T)L(T)\Delta t + q_{k+1}(T)L(T)\Delta t}{1-Q(T)\Delta t}.
\end{aligned} \tag{98}$$

Obviously, $q_0(T)=0$. So we can organize the above three cases into the same expression:

$$p_k(T+\Delta t) \approx \frac{p(k) - p(k)Q(T)\Delta t - q_k(T)L(T)\Delta t + q_{k+1}(T)L(T)\Delta t}{1-Q(T)\Delta t}. \tag{99}$$

Appendix C. Tables

Table 1
Degree distribution of a scale-free network with average degree 11.7.

degree	6	7	8	9	10	11	12
degree distribution	0.2600	0.1330	0.1070	0.0800	0.0600	0.0533	0.0267
degree	13	14	15	16	17	18	19
degree distribution	0.0533	0.0267	0.0200	0.0133	0.0200	0.0000	0.0067
degree	20	21	22	23	24	25	26
degree distribution	0.0067	0.0067	0.0067	0.0200	0.0000	0.0067	0.0133
degree	27	28	29	30	31	32	33
degree distribution	0.0133	0.0133	0.0000	0.0000	0.0000	0.0133	0.0000
degree	34	35	36	37	38	39	40
degree distribution	0.0000	0.0133	0.0000	0.0000	0.0000	0.0067	0.0067
degree	41	42	43	44			
degree distribution	0.0067	0.0000	0.0000	0.0067			

Table 2
Degree distribution on November 29. (All data are rounded.)

degree	6	7	8	9	10	11	12
degree distribution	0.2600	0.1333	0.1067	0.0800	0.0600	0.0533	0.0267
degree	13	14	15	16	17	18	19
degree distribution	0.0533	0.0267	0.0200	0.0133	0.0200	0.0000	0.0067
degree	20	21	22	23	24	25	26
degree distribution	0.0067	0.0067	0.0067	0.0200	0.0000	0.0067	0.0133
degree	27	28	29	30	31	32	33
degree distribution	0.0133	0.0133	0.0000	0.0000	0.0000	0.0133	0.0000
degree	34	35	36	37	38	39	40
degree distribution	0.0000	0.0133	0.0000	0.0000	0.0000	0.0067	0.0067
degree	41	42	43	44			
degree distribution	0.0067	0.0000	0.0000	0.0067			

References

Ball, F. G., Knock, E. S., & O'Neill, P. D. (2011). Threshold behavior of emerging epidemics featuring contact tracing. *Advances in Applied Probability, 43*(4), 1048–1065. https://doi.org/10.1239/aap/1324045698

Ball, F. G., O'Neill, P. D., & Pike, J. (2007). Stochastic epidemic models in structured populations featuring dynamic vaccination and isolation. *Journal of Applied Probability, 44*(3), 571–585. https://doi.org/10.1239/jap/1189717530

Ball, F. G., O'Neill, P. D., & Pike, J. (2008). Control of emerging infectious diseases using responsive imperfect vaccination and isolation. *Mathematical Biosciences, 216*(1), 100–113. https://doi.org/10.1016/j.mbs.2008.08.008

Barabási, A. R. (1999). Emergence of scaling in random networks. *Science, 286*(5439), 509–512. https://www.science.org/doi/abs/10.1126/science.286.5439.509.

Boccaletti, S., Latora, V., Moreno, Y., Chavez, M., & Hwang, D. U. (2006). Complex networks: Structure and dynamics. *Physics Reports*. https://doi.org/10.1016/j.physrep.2005.10.009

Boguná, M., & Pastor-Satorras, R. (2002). Epidemic spreading in correlated complex networks. *Physical Review E - Statistical Physics, Plasmas, Fluids, and Related Interdisciplinary Topics, 66*(4), Article 047104. https://doi.org/10.1103/PhysRevE.66.047104

Brockmann, D., Hufnagel, L., & Geisel, T. (2006). The scaling laws of human travel. *Nature, 439*(7075), 462–465. https://doi.org/10.1038/nature04292

Budnick, B., Biham, O., & Katzav, E. (2022). Structure of networks that evolve under a combination of growth and contraction. *Physical Review E - Statistical Physics, Plasmas, Fluids, and Related Interdisciplinary Topics, 106*(4), Article 044305. https://doi.org/10.1103/PhysRevE.106.044305

Budnick, B., Biham, O., & Katzav, E. (2025). Phase transition in evolving networks that combine preferential attachment and random node deletion. *J Stat Mech-theory E, 2025*(1), Article 013401. https://iopscience.iop.org/article/10.1088/1742-5468/ad99c7/meta.

Chen, B., Zhao, Y., Jin, Z., He, D., & Li, H. (2023). Twice evasions of Omicron variants explain the temporal patterns in six Asian and Oceanic countries. *BMC Infectious Diseases, 23*(1), 25. https://doi.org/10.1186/s12879-023-07984-9

Chen, X., Yang, X., Wang, R., Li, A., Yang, X., Cai, S., & Wang, W. (2025). Nonlinear effects of resource allocation delay on epidemic spreading in complex networks. *Chaos, 35*(1). https://doi.org/10.1063/5.0227075

Chen, X., Zhang, Y. C., Zhang, J., Zhou, M., & He, Q. (2023). Detoxification characteristics of patients with novel coronavirus infection in different periods. *Chinese Tropical Medicine, 23*(3), 310–313.

COVID-19 prevention and control program (seventh edition). (2020). *Chinese Journal of Infection Control, 19*(11), 1042–1048.

Endo, A., Leclerc, Q. J., Knight, G. M., Medley, G. F., Atkins, K. E., Funk, S., & Kucharski, A. J. (2020). Implication of backward contact tracing in the presence of overdispersed transmission in COVID-19 outbreaks. *Wellcome Open Res, 5*, 239. https://doi.org/10.12688/wellcomeopenres.16344.3

Gamerman, D., & Lopes, H. F. (2006). *Markov chain Monte Carlo: Stochastic simulation for bayesian inference*. CRC Press. https://doi.org/10.1201/9781482296426

Gonzalez, M. C., Hidalgo, C. A., & Barabasi, A. L. (2008). Understanding individual human mobility patterns. *Nature, 453*(7196), 779–782. https://doi.org/10.1038/nature06958

Haario, H., Laine, M., Mira, A., & Saksman, E. (2006). DRAM: Efficient adaptive MCMC. *Statistics and Computing, 16*(4), 339–354. https://doi.org/10.1007/s11222-006-9438-0

Jin, Z., Sun, G. Q., & Liu, M. X. (2014). *Modelling and analysis of epidemic spreading on networks*. Science Press.

Jordan, D., & Smith, P. (2007). *Nonlinear ordinary differential equations* (4th ed.). OUP Oxford.

Kanji, J. N., Zelyas, N., MacDonald, C., Pabbaraju, K., Khan, M. N., Prasad, A., Hu, J., Diggle, M., Berenger, B. M., & Tipples, G. (2023). False negative rate of COVID-19 pcr testing: A discordant testing analysis. *Virology Journal, 18*(1), 1–6. https://doi.org/10.1186/s12985-021-01489-0

Klafter, J., Shlesinger, M. F., & Zumofen, G. (1996). Beyond brownian motion. *Physics Today, 49*(2), 33–39. https://doi.org/10.1063/1.881487

Kojaku, S., Hébert-Dufresne, L., Mones, E., Lehmann, S., & Ahn, Y. Y. (2021). The effectiveness of backward contact tracing in networks. *Nature Physics, 17*(5), 652–658. https://doi.org/10.1038/s41567-021-01187-2

Li, Q., Xiao, Y. N., Wu, J. H., & Tang, S. Y. (2020). COVID-19 epidemic time-lag model construction and confirmed case-driven tracking and isolation measures analysis. *Acta Mathematical Applagatae Sinica, 43*(2), 13.

Luo, X. F., & Jin, Z. (2020). A new insight into isolating the high-degree nodes in network to control infectious diseases. *Commun Nonlinear Sci, 91*, Article 105363. https://doi.org/10.1016/j.cnsns.2020.105363

Ma, Z. E., Zhou, Y. C., Wang, W. D., & Jin, Z. (2004). *Mathematical modeling and research of infectious disease dynamics*. Science Press.

Mielke, A., & Christiansen, L. E. (2025). Duration of transients in outbreaks: When can infectiousness be estimated? *Journal of Mathematical Biology, 90*(1), 11. https://doi.org/10.1007/s00285-024-02175-9

Miller, J. C. (2011). A note on a paper by erik Volz: SIR dynamics in random networks. *Journal of Mathematical Biology, 62*(3), 349–358. https://doi.org/10.1007/s00285-010-0337-9

Milo, R., Itzkovitz, S., Kashtan, N., Levitt, R., Shen-Orr, S., Ayzenshtat, I., Sheffer, M., & Alon, U. (2004). Superfamilies of evolved and designed networks. *Science, 303*(5663), 1538–1542. https://doi.org/10.1126/science.1089167

Moreno, Y., Pastor-Satorras, R., & Vespignani, A. (2002). Epidemic outbreaks in complex heterogeneous networks. *The European Physical Journal B*, (26), 521–529. https://doi.org/10.1140/epjb/e20020122

Mossong, J., Hens, N., Jit, M., Beutels, P., Auranen, K., Mikolajczyk, R., … Edmunds, W. J. (2008). Social contacts and mixing patterns relevant to the spread of infectious diseases. *PLoS medicine, 5*(3), e74. https://doi.org/10.1371/journal.pmed.0050074

Nava, A., Papa, A., Rossi, M., & Giuliano, D. (2020). Analytical and cellular automaton approach to a generalized SEIR model for infection spread in an open crowded space. *Physical Review Research, 2*(4), Article 043379. https://doi.org/10.1103/PhysRevResearch.2.043379

Newman, M. E. J. (2002). Spread of epidemic disease on networks. *Physical Review E - Statistical Physics, Plasmas, Fluids, and Related Interdisciplinary Topics, 66*(1), Article 016128. https://doi.org/10.1103/PhysRevE.66.016128

Newman, M. E. J. (2009). Random graphs with clustering. *Physical Review Letters, 103*(5), Article 058701. https://link.aps.org/doi/10.1103/PhysRevLett.103.058701.

Newman, M. E. J. (2010). *Networks: An introduction*. Oxford university press.

Newman, M. E. J., Strogatz, S. H., & Watts, D. J. (2001). Random graphs with arbitrary degree distributions and their applications. *Physical Review E - Statistical Physics, Plasmas, Fluids, and Related Interdisciplinary Topics, 64*(2), Article 026118. https://link.aps.org/doi/10.1103/PhysRevE.64.026118.

Okolie, A., & Müller, J. (2020). Exact and approximate formulas for contact tracing on random trees. *Mathematical Biosciences, 321*, Article 108320. https://doi.org/10.1016/j.mbs.2020.108320

[data set] Omicron strains data in UK from PHE. Public Health England. https://ukhsa-dashboard.data.gov.uk/respiratory-viruses/covid-19cases.

Pastor-Satorras, R., & Vespignani, A. (2001a). Epidemic spreading in scale-free networks. *Physical Review Letters, 86*(14), 3200–3203. https://doi.org/10.1103/PhysRevLett.86.3200

Pastor-Satorras, R., & Vespignani, A. (2001b). Epidemic dynamics and endemic states in complex networks. *Physical Review E - Statistical Physics, Plasmas, Fluids, and Related Interdisciplinary Topics, 63*(6), Article 066117. https://doi.org/10.1103/PhysRevE.63.066117

Pastor-Satorras, R., & Vespignani, A. (2002). Epidemic dynamics in finite size scale-free networks. *Physical Review E - Statistical Physics, Plasmas, Fluids, and Related Interdisciplinary Topics, 65*(3), Article 035108. https://doi.org/10.1103/PhysRevE.65.035108

Pellis, L., Ball, F., Bansal, S., Eames, K., House, T., Isham, V., & Trapman, P. (2015). Eight challenges for network epidemic models. *Epidemics-Neth, 10*, 58–62. https://doi.org/10.1016/j.epidem.2014.07.003

Prem, K., Cook, A. R., & Jit, M. (2017). Projecting social contact matrices in 152 countries using contact surveys and demographic data. *PLoS Computational Biology, 13*(9), Article e1005697. https://doi.org/10.1371/journal.pcbi.1005697

Sabbatino, M., De Reggi, S., & Pugliese, A. (2025). A theoretical analysis of mass testing strategies to control epidemics. *Bulletin of Mathematical Biology, 87*(2), 22. https://doi.org/10.1007/s11538-024-01387-W

Song, C., Havlin, S., & Makse, H. A. (2005). Self-similarity of complex networks. *Nature, 433*(7024), 392–395. https://doi.org/10.1038/nature03248

Strogatz, S. H. (2001). Exploring complex networks. *Nature, 410*(6825), 268–276. https://doi.org/10.1038/35065725

Szabaó, G., & Fath, G. (2007). Evolutionary games on graphs. *Physics Reports, 446*(4–6), 97–216. https://doi.org/10.1016/j.physrep.2007.04.004

Szolnoki, A., & Perc, M. (2009). Resolving social dilemmas on evolving random networks. *EPL-Europhys Lett, 86*(3), Article 30007. https://doi.org/10.1209/0295-5075/86/30007

Szolnoki, A., Perc, M., & Danku, Z. (2008). Making new connections towards cooperation in the Prisoner's Dilemma game. *EPL-Europhys Lett, 84*(5), Article 50007. https://doi.org/10.1209/0295-5075/84/50007

Tishby, I., Biham, O., & Katzav, E. (2019). Convergence towards an Erdős-Rényi graph structure in network contraction processes. *Physical Review E - Statistical Physics, Plasmas, Fluids, and Related Interdisciplinary Topics, 100*(3), Article 032314. https://doi.org/10.1103/PhysRevE.100.032314

Tishby, I., Biham, O., & Katzav, E. (2020). Analysis of the convergence of the degree distribution of contracting random networks towards a Poisson distribution using the relative entropy. *Physical Review E - Statistical Physics, Plasmas, Fluids, and Related Interdisciplinary Topics, 101*(6), Article 062308. https://doi.org/10.1103/PhysRevE.101.062308

Viswanathan, G. M., Afanasyev, V., Buldyrev, S. V., Murphy, E. J., Prince, P. A., & Stanley, H. E. (1996). Lévy flight search patterns of wandering albatrosses. *Nature, 381*(6581), 413–415. https://doi.org/10.1038/381413a0

Volz, E. (2008). SIR dynamics in random networks with heterogeneous connectivity. *Journal of Mathematical Biology, 56*, 293–310. https://doi.org/10.1007/s00285-007-0116-4

Wen, Z., Chi, Y., Zhang, L., Liu, H., Du, K., Li, Z., Chen, J., Cheng, L., & Wang, D. (2020). Coronavirus disease 2019: initial detection on chest CT in a retrospective multicenter study of 103 Chinese patients. *Radiology: Cardiothoracic Imaging, 2*(2), Article e200092. https://doi.org/10.1148/ryct.2020200092

Xiao, Y. N., Zhou, Y. C., & Tang, S. Y. (2012). *The principle of biological mathematics*. Xi 'an Jiaotong University Press.

Zhang, H., Cao, L., Fu, C., Cai, S., & Gao, Y. (2023). Epidemic spreading on multi-layer networks with active nodes. *Chaos, 33*(7). https://doi.org/10.1063/5.0151777

Zhang, S., Jin, Z., & Zhang, J. (2018). The dynamical modeling analysis of the spreading of passive worms in P2P networks. *Discrete Dynamics in Nature and Society, 2018*(1), Article 1656907. https://doi.org/10.1155/2018/1656907

Zhang, Y., Xie, X. J., & Yu, X. D. (2022). Analysis of epidemiological characteristics of patients infected with COVID-19 Omicron variant strain in a cabin hospital in Shanghai. *Infectious Diseases Information, 35*(4), 6.